DERMATOSURGRY 2

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SCE DERMATOLOGY / EBDV BY DR :EMAN GOMAA - CONNECT

What are flaps?

Tissues that are moved from one place to another, and carry their own vascularity with them

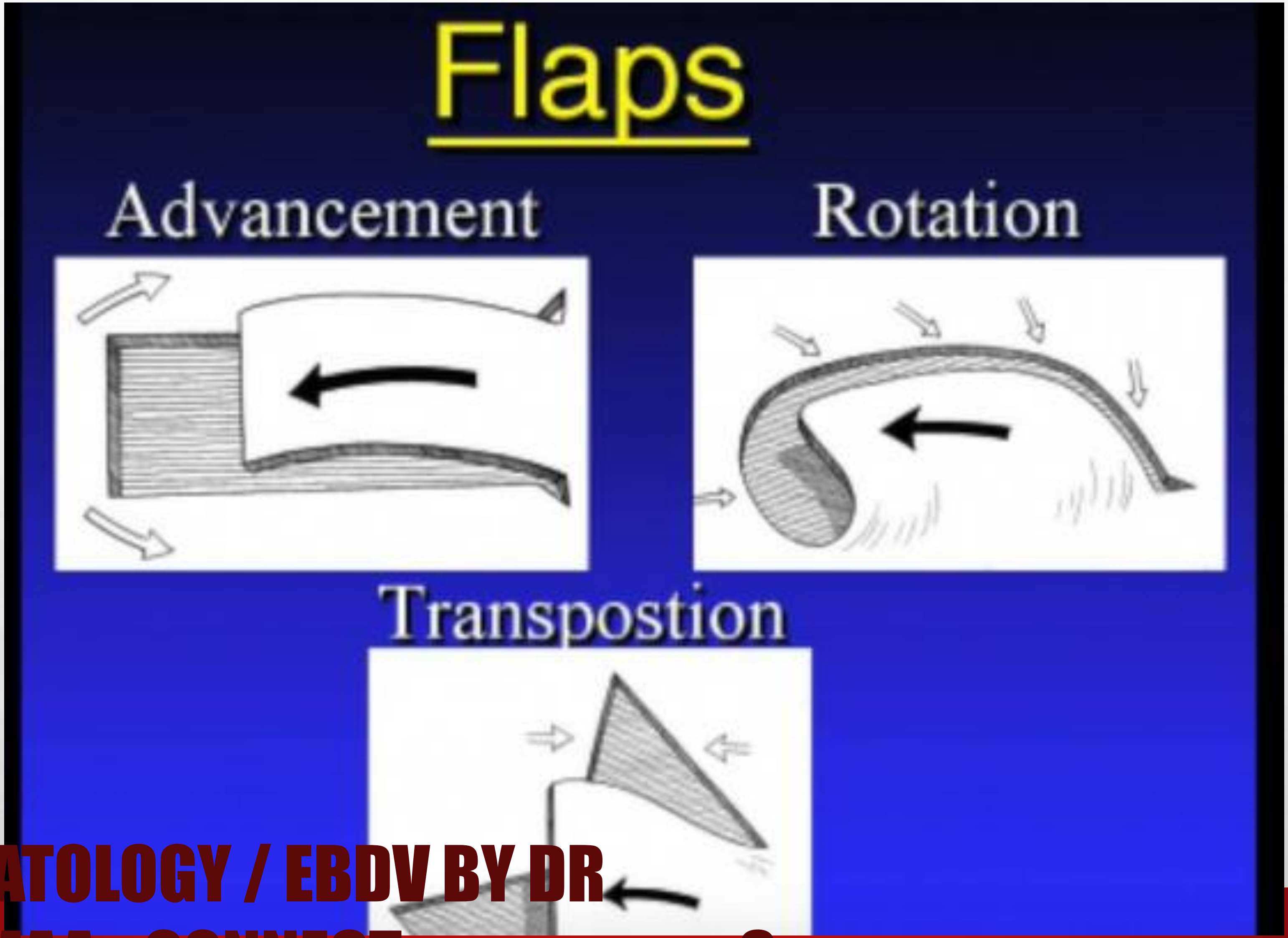
Advantages of flap cover

- Have their own blood supply
- Have compatible tissue characteristics
- Lesser chance of contractures
- Protection of underlying structures
- Durability

5 steps of flap reconstructive surgery

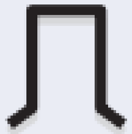
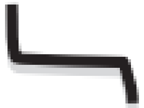
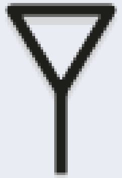
- Selection
- Siting
- Designing
- Construction
- Transfer of flap

- Transfer of flap consists of primary and secondary movements
- The primary movement is the movement of flap
- The secondary movement is the movement of surrounding tissue (To close the secondary defect)

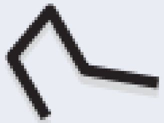
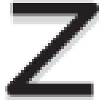

Flaps
Advancement
Rotation
Transpostion

Local flaps

- Transposition flaps
- Rotation flaps
- Z plasty
- Rhomboid flaps
- Advancement flaps
- Keystone flaps

Type of Flap	Description	Appearance
Advancement Flap		
Unidirectional, uncomplicated advancement of leading edge of flap		
Unilateral advancement flap (U-plasty)	Defect excised as square and incision extended in same direction on two but opposite parallel sides of defect; burow's triangle created at end of each extension and flap slides over defect creating U-shaped scar	
Bilateral advancement flap (H-plasty)	Double U-plasty or double advancement flap; two U-plasty flaps created as mirror images of one another; most useful for scalp and eyebrow defects (H-plasty)	H
Bilateral T-plasty (A-T, O-T)	Linear repair of wound perpendicular to pre-existing cosmetic boundary; useful for above brow, upper cutaneous lateral lip	T
Burow's advancement flap	Defect excised in shape of equilateral triangle and one arm of triangle extended; burow's triangle created at contralateral side of extension and tissue slides to cover defect	
Island pedicle flap	Special advancement flap: most of vascular supply from a subcutaneous pedicle (remains attached to central portion of flap) and all dermal margins of flap severed before advanced	

Rotation flap		
Recruits adjacent tissue laxity and directs wound tension vectors away from primary surgical defect; curvilinear incision (arc) adjacent to primary defect and flap rotated to primary defect site; useful for scalp, temple, and medial cheek defects		
Dorsal nasal rotation flap	Special type of rotation flap; long sweeping arc that involves rotation of entire nasal dorsum (elevated at level of perichondrium or periosteum)	J
Bilateral advancement rotational flap (O – Z flap)	Bilateral rotation flap converting circular defect into a Z-shaped incision line, most useful on scalp (can be purely rotational or advancement with rotation)	Z

Type of Flap	Description	Appearance
Transposition Flap		
Most complex design, redirects wound closure tension, moves tissue from area of surplus to area of need by **transpositioning across intervening islands of unaffected tissue**		
Rhomboid transposition flap	Rhomboidal-shaped flap created adjacent to round or oval defect and transposed into defect	
Bilobed transposition flap	Recruits tissue from proximal nasal dorsum (more laxity) and transfers to defect, useful on distal nose	
Nasolabial transposition flap	Flap from medial cheek adjacent to melolabial fold transposed to alar wound, useful in lateral and central alar wounds	
Z-plasty	Useful for scars crossing relaxed tension lines or releasing contractures (redistributes tension over wound)	
Paramedian forehead flap	2 stage flap for repair of subtotal to total nasal defects; forehead flap designed vertically to preserve supratrochlear artery supply; flap rotated 180 degrees and sewn into nasal defect; 2-3 weeks later pedicle divided and repositioned	**Axial pattern flap** as well

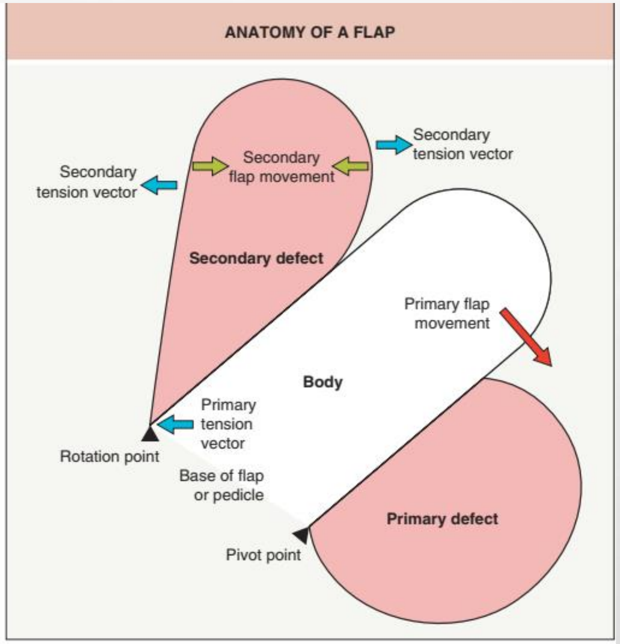


Fig. 147.3 Anatomy of a flap. Definitions of the components of a flap are outlined in Table 147.2.

DEFINITIONS OF FLAP COMPONENTS	
Body	The skin being advanced, transposed, rotated, interpolated or imported into the primary defect
Pedicle	Also known as the vascular base of the flap, it is the conduit of the vascular supply that will remain intact and maintain the vascularization of the body of the flap intraoperatively and for the early postoperative period
Primary defect	The area deficient in skin that will be reconstructed by the movement of the body of the flap
Secondary defect (donor site defect)	The area devoid of skin created by the movement of the body of the flap
Primary flap movement	The movement of the flap body into the defect
Secondary flap movement	The tissue movement necessary to close the void or donor site defect created by the movement of the body of the flap
Primary tension vector	The direction of the force tending to counteract the movement of the body of the flap
Secondary tension vector	The direction of the force created by the closure of the donor site defect

CLASSIFICATION OF FLAPS BASED ON DESIGN CHARACTERISTICS					
Flap category	**Flap variations**	**Synonyms**	**Mechanism of tissue movement**	**Advantages**	**Disadvantages**
Burow's triangle displacement flaps	Single tangent adv flap	Burow's flap	• Tissue advancement: dependent upon intrinsic tissue elasticity	• Enables closures adjacent to vital structures through displacement of Burow's triangles • Single tangent flaps have excellent blood supply	• Depends on intrinsic skin laxity of the flap skin • Requires extensive undermining • Double tangent flaps have a more limited blood supply
	Bilateral single tangent adv flap	A-to-T flap, O-to-T flap			
	Double tangent adv flap	U-flap			
	Bilateral double tangent adv flap	H-flap			
	Curvilinear tangent adv flap	Rotation flap, Karapandzic flap, Mustarde flap			
Defect reconfiguration flaps	Island pedicle flap	Kite flap, myocutaneous pedicle flap	• Tissue advancement: dependent upon intrinsic elasticity of subcutaneous pedicle • Tissue movement results in reconfiguration of the defect	• Tissue-conservative due to reconfiguration of the defect without removal of Burow's triangle	• Blood supply is limited to the subcutaneous pedicle • Use limited to locations with spongy, elastic subcutaneous tissue
Tissue reorientation flaps	Rhombic transposition flap	Limberg flap, Dufourmentel flap, Webster's 30° flap	• Reorientation of adjacent tissue laxity	• Effective for tight closures • Directional tissue gain can be achieved in direction of tissue deficit	• Depends on laxity adjacent to the defect • Multidirectional incisions; difficult to place all in RSTL
	Bilobed transposition flap				
	Nasolabial transposition flap	Melolabial fold flap			
	Nasolabial flap with a twist (Spear's flap)				
Tissue importation flaps	Forehead flap	Indian flap	• Importation of tissue from a distant site	• Coverage of large wounds • Can provide vascularity for coverage of avascular grafts such as cartilage grafts	• Two-stage flap • Vascular supply limited to narrow pedicle
	Nasolabial interpolation flap	Melolabial interpolation flap			
	Retroauricular pedicled flap	Pin back flap			
	Modified Hughes flap Abbe cross-lip flap				

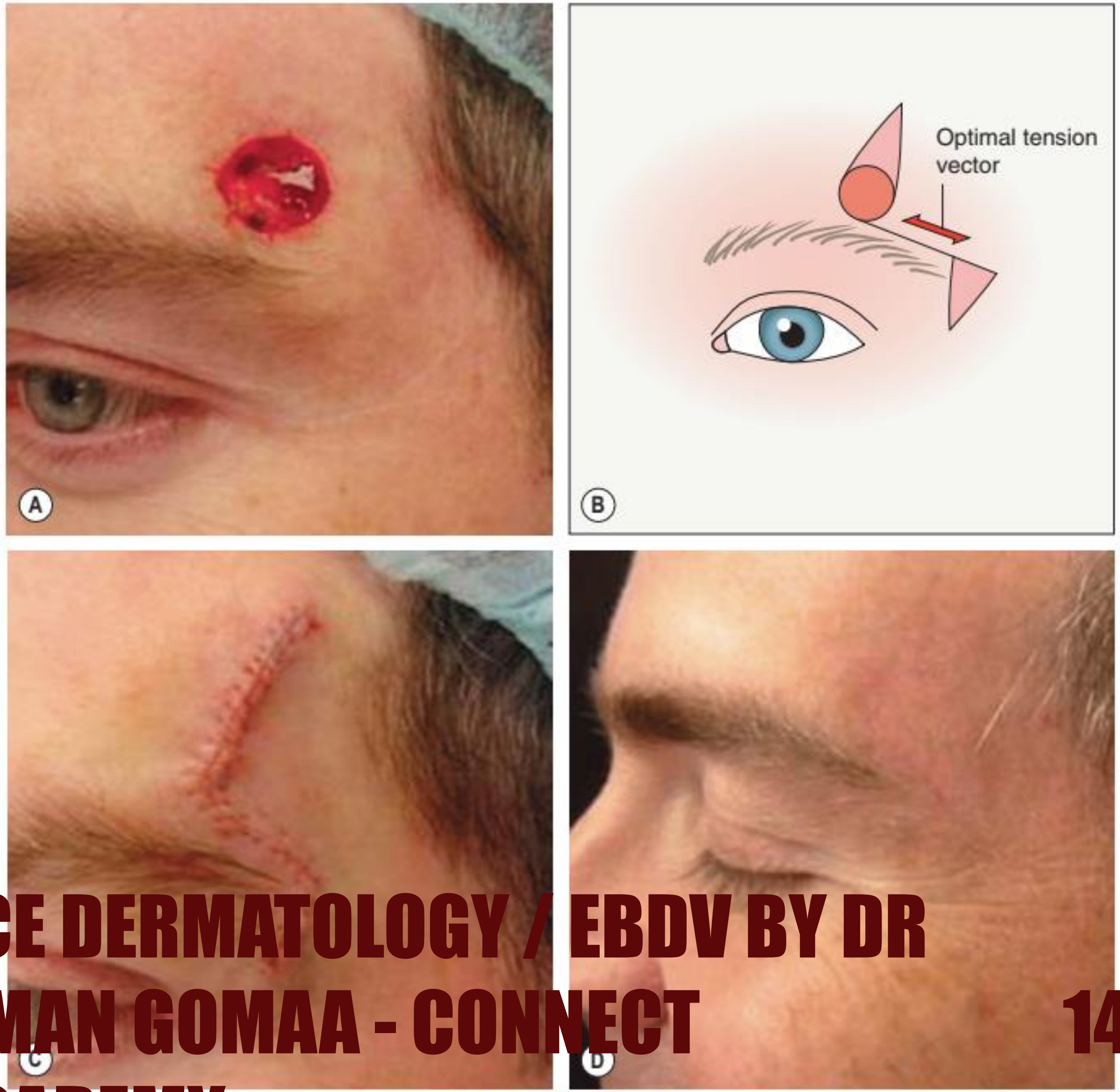
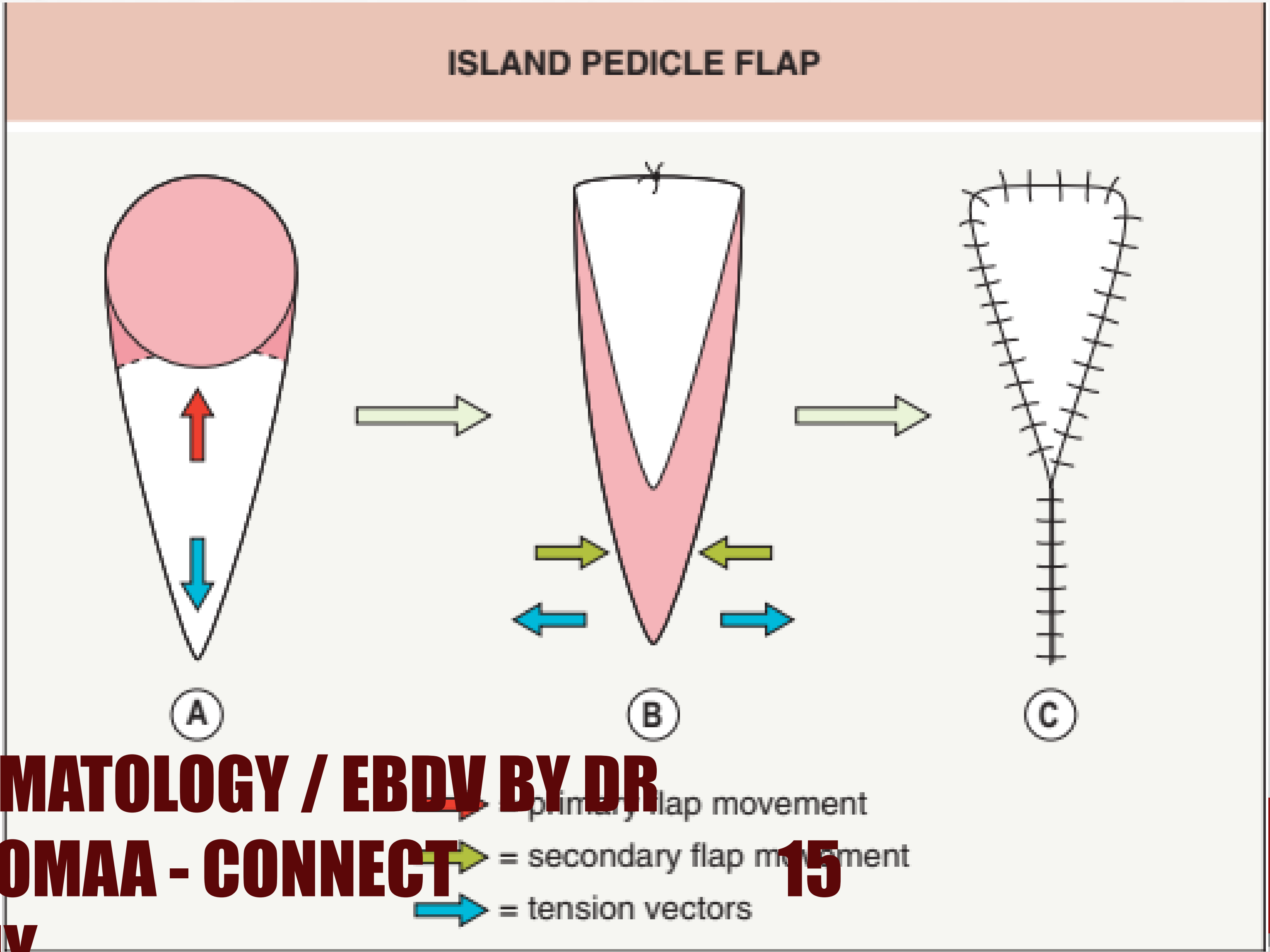


Fig. 147.5 Single tangent advancement flap (Burow's flap). A The wound extends to the brow. **B** Determine the tension vector that will cause the least distortion and make the tangential incision parallel to this optimal tension vector. Both Burow's triangles (displaced and adjacent to defect) are oriented perpendicular to the optimal tension vector. **C** Closure with a single tangent advancement flap with the tangent extending lateral to the brow and displacing a Burow's triangle to the temple and lateral orbit. **D** The eyebrow is entirely preserved.


ISLAND PEDICLE FLAP
A
B
C
flap movement
= secondary flap
= tension vectors

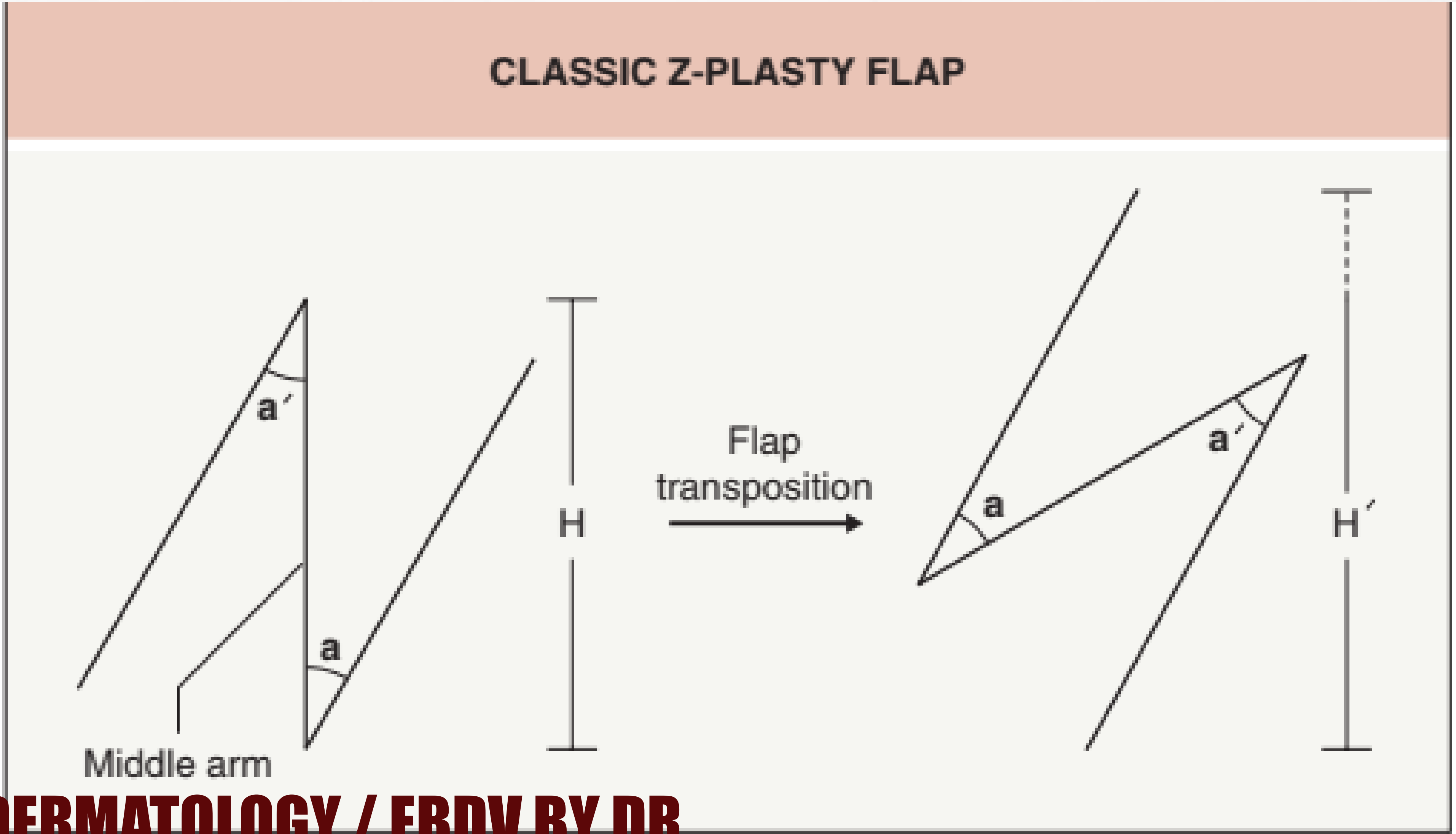


Fig. 147.7 Classic Z-plasty flap. The directional tissue gain (dotted line) is depicted (H to H´) after flap transposition in the direction of the middle arm of the Z-plasty.

Fig. 147.8 Classic Z-plasty flap. A Defect close to the eyelid margin. The middle arm of this Z-plasty flap is composed of the defect *plus* the Burow's triangles excised above and below it to facilitate the transposition of the flaps (*). The medial flap is transposed over the lateral flap (arrows). **B** The wound is closed via a Z-plasty closure with the transposition of flaps at the upper medial and lower lateral aspects of the wound. This is done to lengthen tissue in the direction of the palpebral margin in order to prevent ectropion as demonstrated at the 4-month postoperative visit (**C**).

Fig. 147.10 Bilobed flap. A Defect on the nasal tip adjacent to the free margin of the ala. Bilobed flap reconstruction with minimal distortion of the alar rim (**B**) and long-term result (**C**).

Fig. 147.11 Rotation flap. Large defect (**A**) on the medial cheek and lower eyelid reconstructed with a rotation flap (Mustarde flap) with displacement of the superior Burow's triangle laterally along the curvilinear tangent which follows the relaxed skin tension lines (**B**). **C** Final results with no distortion of the eyelid and incision lines nicely camouflaged along cosmetic subunit borders and relaxed skin tension lines.

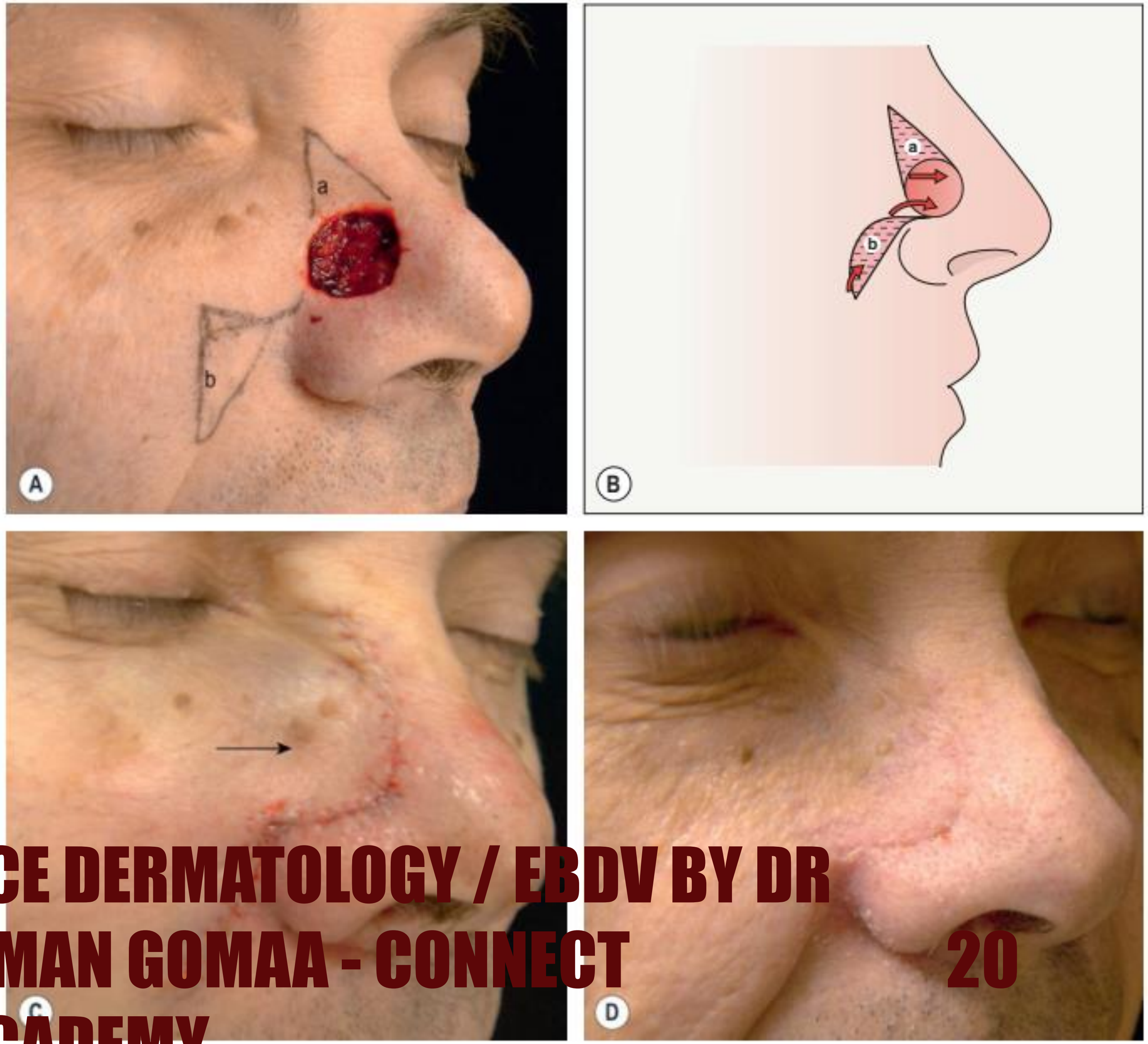
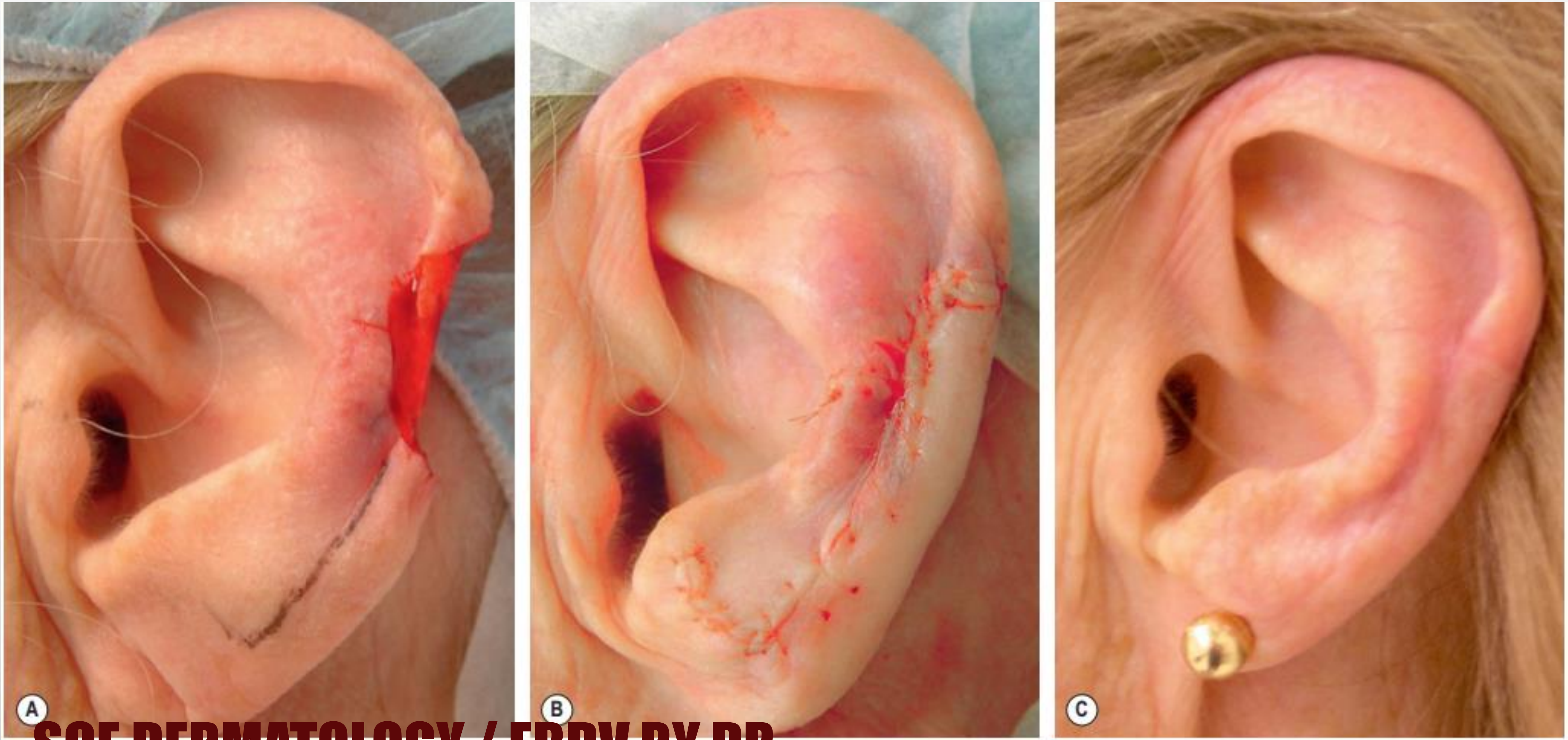


Fig. 147.12 Crescentic advancement flap. A Medium-sized defect on the right inferior nasal sidewall. A single tangent advancement flap (Burow's flap) is planned with a crescent-shaped Burow's triangle on the medial cheek. **B** The crescent (b) fits nicely with the naturally curved anatomic configuration of the alar crease and creates a "wound edges of unequal length" situation which further contributes to the advancement of skin onto the nose. **C** Flap sutured in place; note the subtle indentation (arrow) that corresponds to a periosteal suture placed at the nasofacial sulcus in order to prevent tenting and to recreate the sulcus. **D** Four-month postoperative appearance.

Fig. 147.13 Double tangent advancement flap. A Defect with preoperative plan to advance skin from lateral to the defect, incorporating the residual brow. **B** Sutured flap realigning the brow. **C** Six months postoperatively with reconstitution of the brow. This technique was used to reconstruct and preserve the continuity of the eyebrow in this patient.



Fig[illegible] a small amount of cartilage. A single tangential incision is planned inferiorly within the helical groove to the earlobe. A Burow's triangle excision is planned directly behind the defect on the posterior ear. The flap will be raised by dissection of the ski[illegible] displaced Burow's triangle will be removed from the earlobe and the skin of the flap will be redraped through upward advancement into the defect. B The flap is sutured after a very small wedge of cartilage has been removed to reconstitute the helical rim of cartilage. C Three-month postoperative appearance.

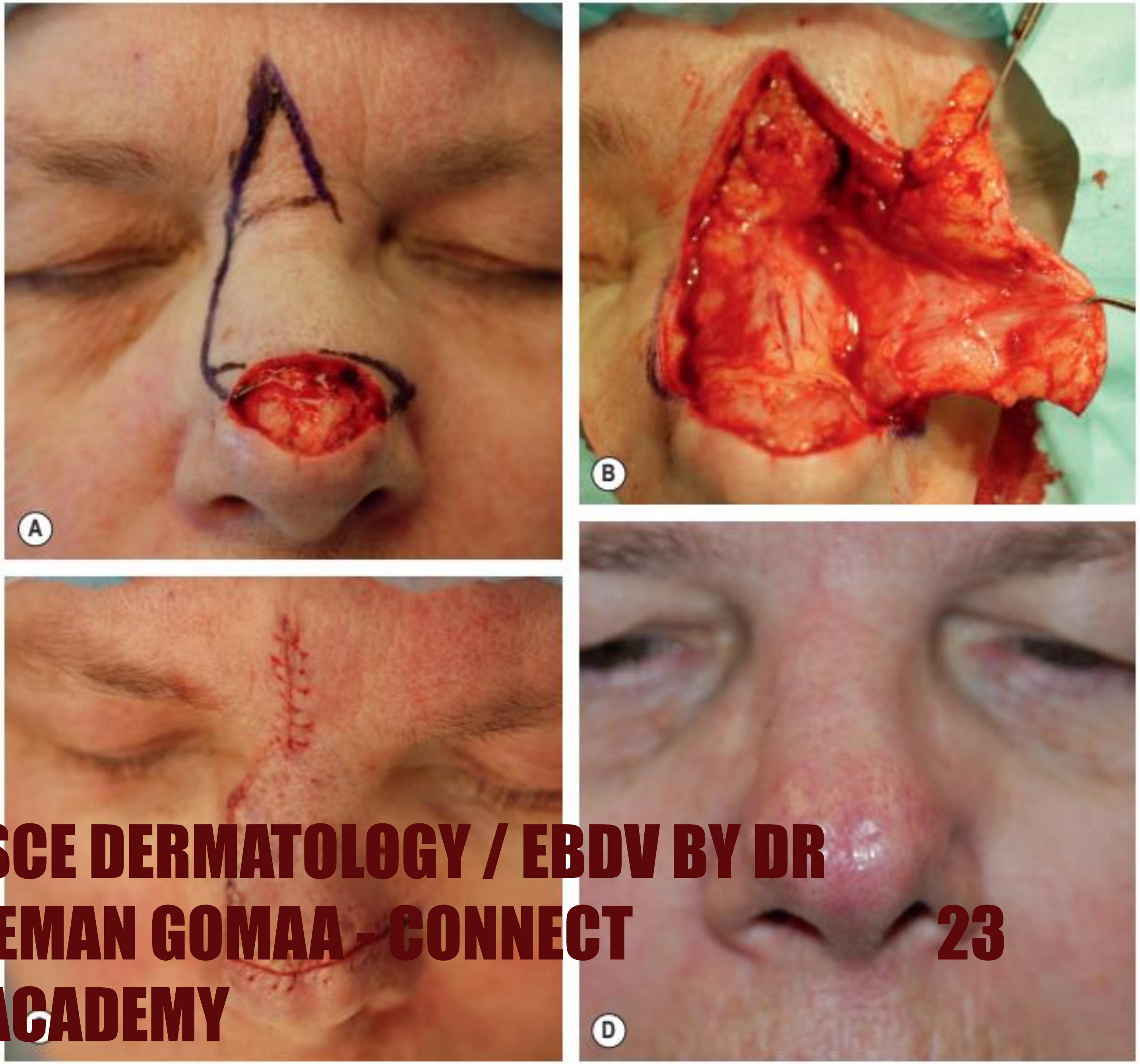


Fig. 147.15 Dorsal nasal flap. A Moderate-sized defect of the nasal tip. **B** The flap is incised and reflected demonstrating the deep plane of dissection just above the nasal periosteum. The flap will be redraped over the tip of the nose with primary closure of the displaced Burow's triangle on the glabella. **C** Flap sutured into place. **D** Result four months postoperatively.

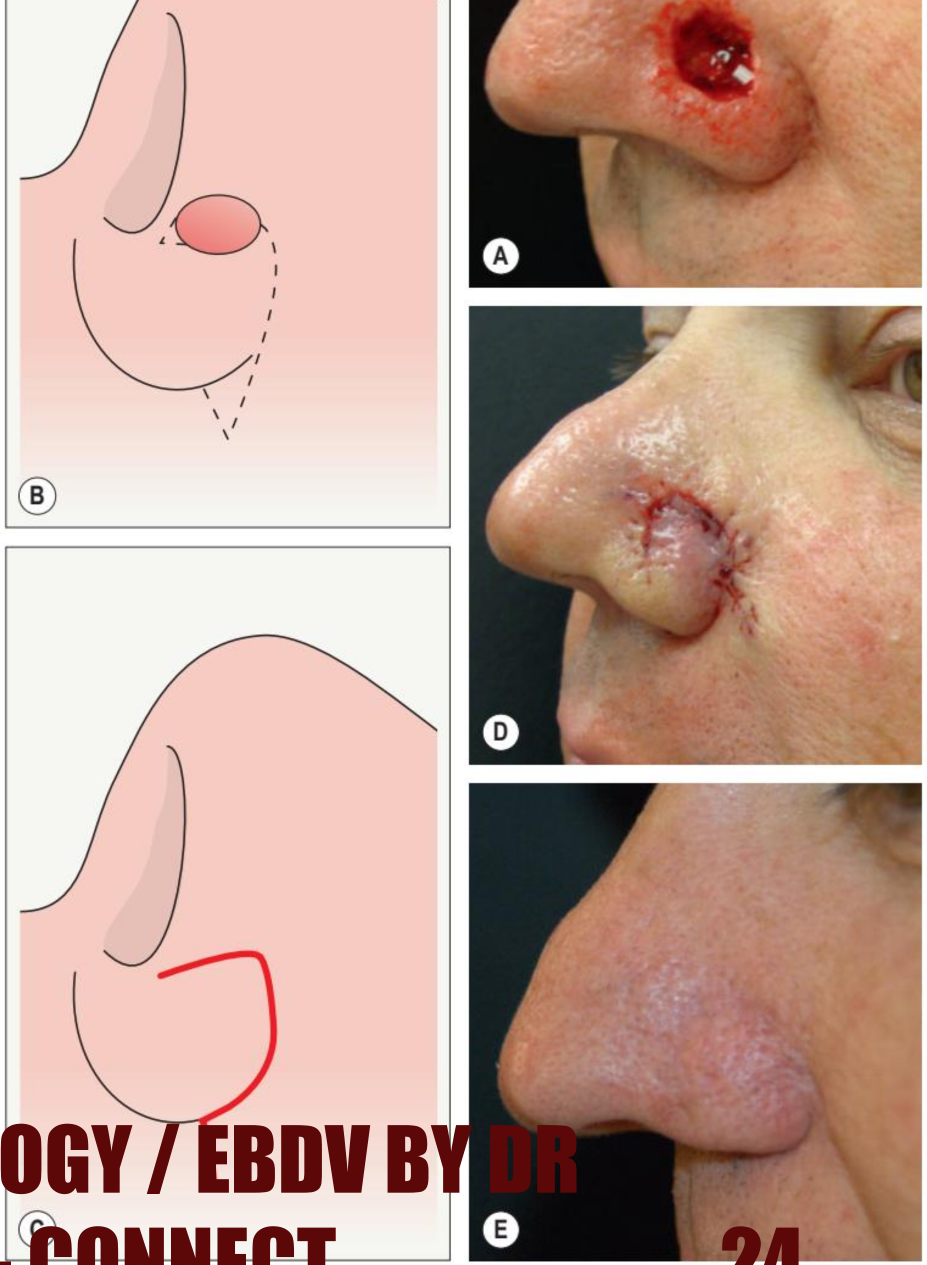


Fig. 14.10 Alar rotation flap (ARF). A These flaps are best suited for mid to medial defects that are <75% of the vertical height of the ala. **B** The design is a rotation flap with a back-cut incised within the alar crease to the apical triangle at the alolabial junction. **C** Closure of the back-cut recreates the crease at the alolabial junction. **D** Incorporation of the back-cut skin onto the ala prevents distortion of the ala. **E** Long-term follow-up.

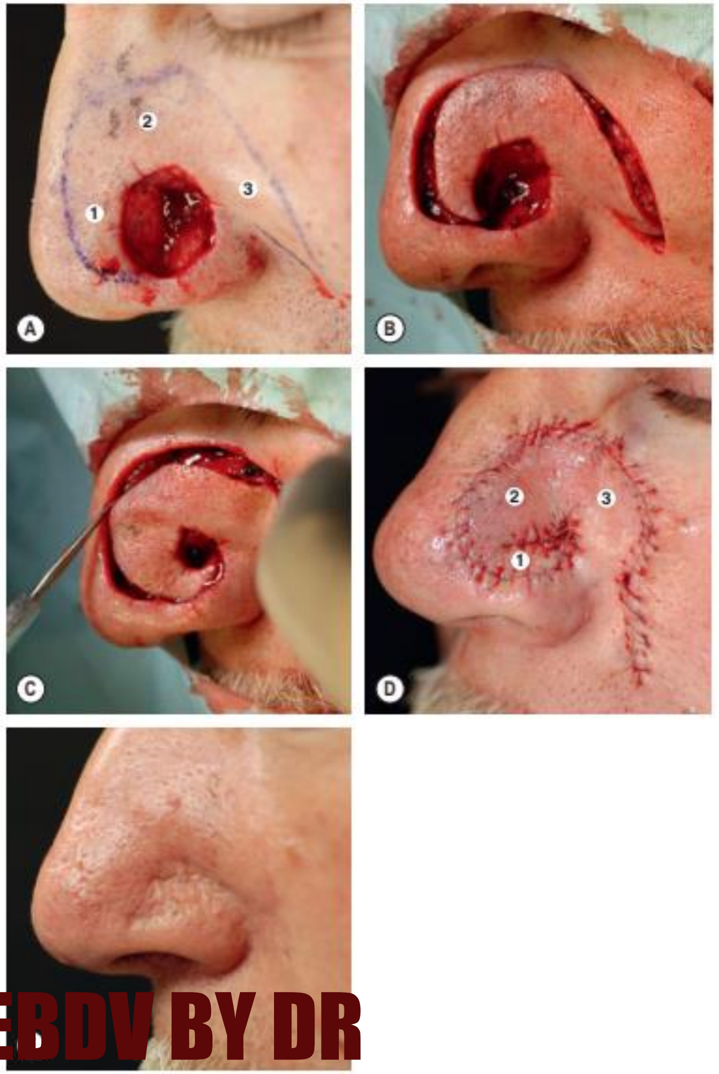


Fig. 147.17 Spiral rotation flap (SRF). **A** These flaps are well suited for larger defects involving the ala, alar crease, and lower nasal sidewall. The flap (outlined in ink) can be conceptualized as having three parts: (1) the alar portion (equals the size of the alar defect); (2) the body; and (3) the tail; the latter two components equal the size of the entire wound. **B** Incision of the flap with a back-cut on the medial cheek along the melolabial junction. **C** After undermining, with the flap base created at the inner aspect of the spiral flap, demonstration of the spiral rotation of the flap into the alar portion of the defect. **D** Sutured flap. **E** Three-month follow-up.

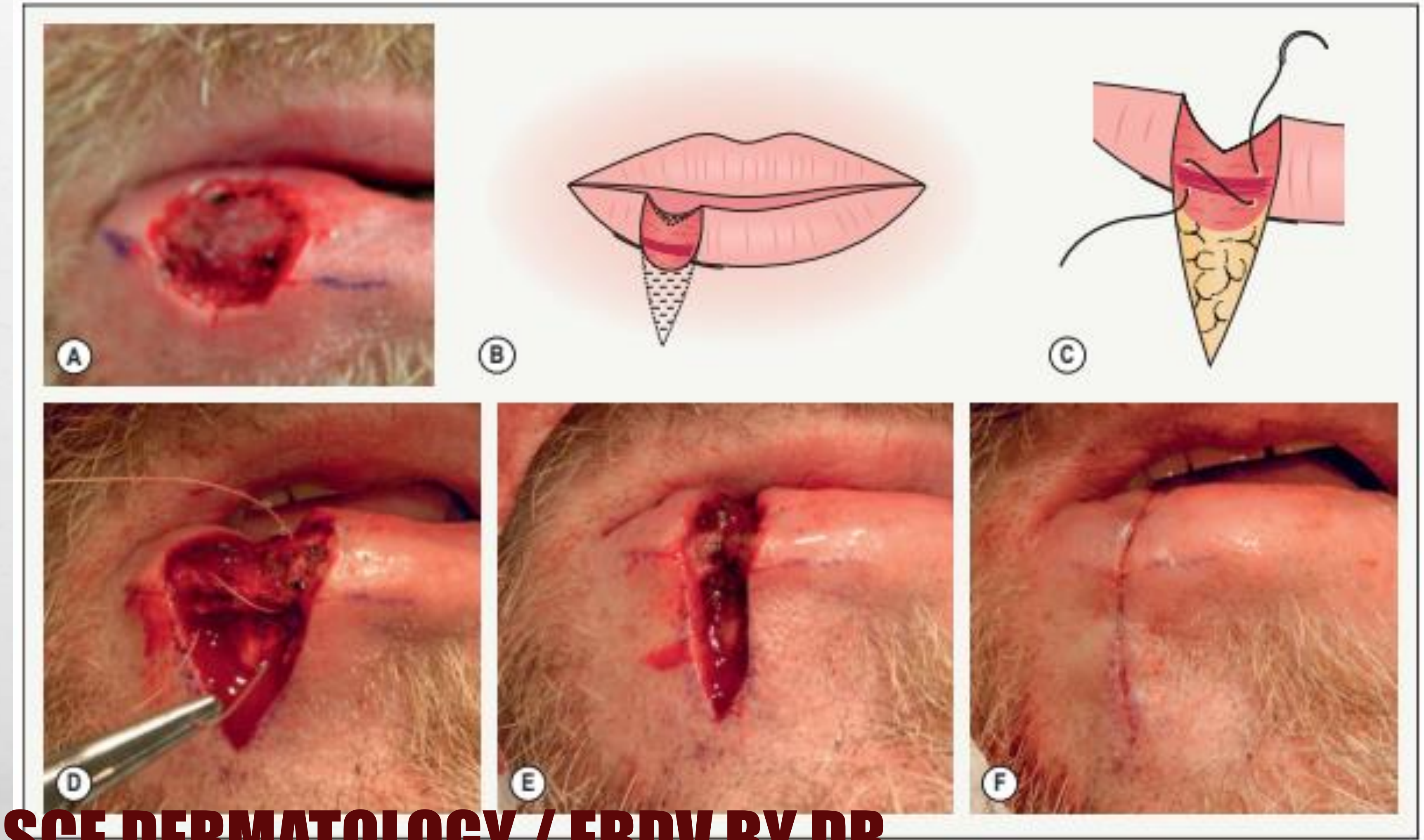


Fig. 147.18 Wedge closure of the lip. A Typical defect after Mohs surgical excision of a squamous cell carcinoma. Note indelible ink lines and nicks in the skin that demarcate the vermilion border, facilitating precise realignment. B Burow's triangles are excised intraorally and extraorally perpendicular to the vermilion border. [illegible] must be removed to avoid excessive redundancy when the wound edges are approximated, an obtuse wedge of muscle is removed. C The anterior margin of the orbicularis oris muscle is identified medially and laterally. A "figure-of-eight" stitch is placed in order to reapproximate the muscle margins. D "Figure-of-eight" suture in place prior to reapproximation of muscle edges. E Realignment of the anterior edge of the [illegible] muscle. F Wound closure prior to the placement of superficial epidermal sutures. Note the near perfect realignment of the vermilion border and the well-proportioned recapitulation of the lower lip.

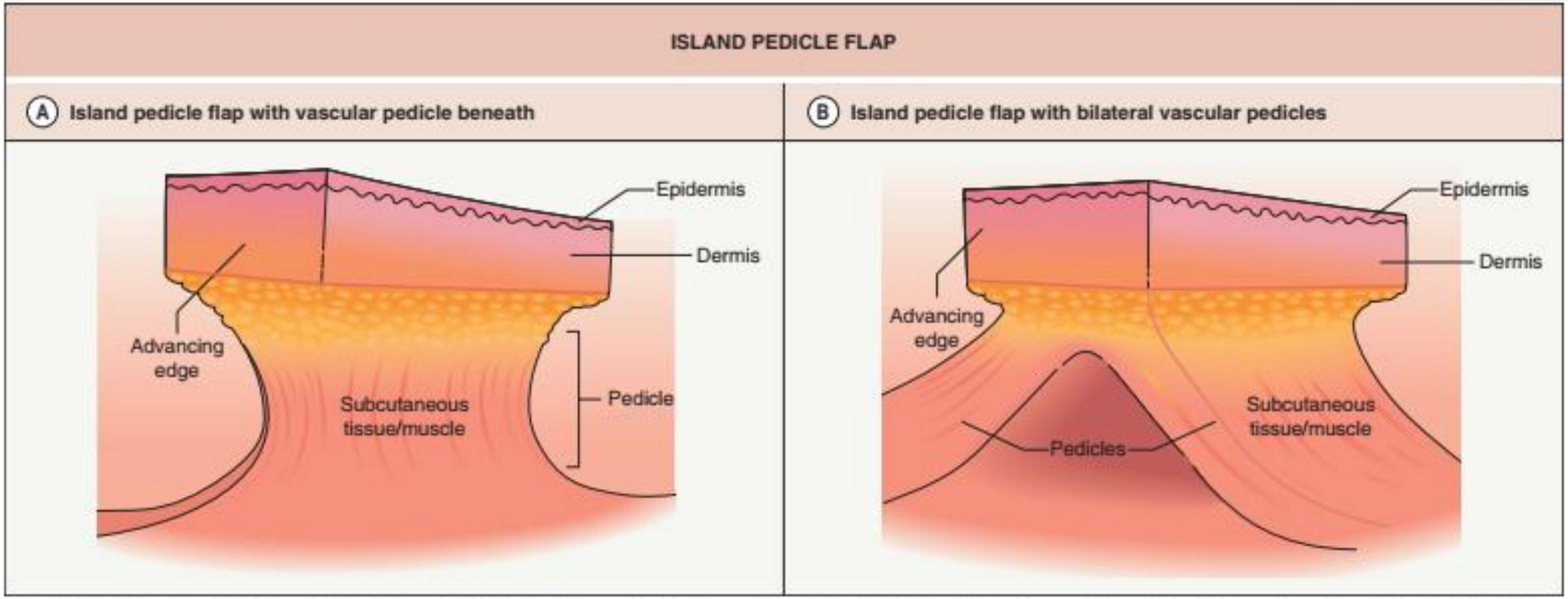


Fig. 147.19 Island pedicle flap. A Vascular pedicle derived immediately beneath the island of skin. **B** Vascular pedicles derived from tissue lateral to the island flap.

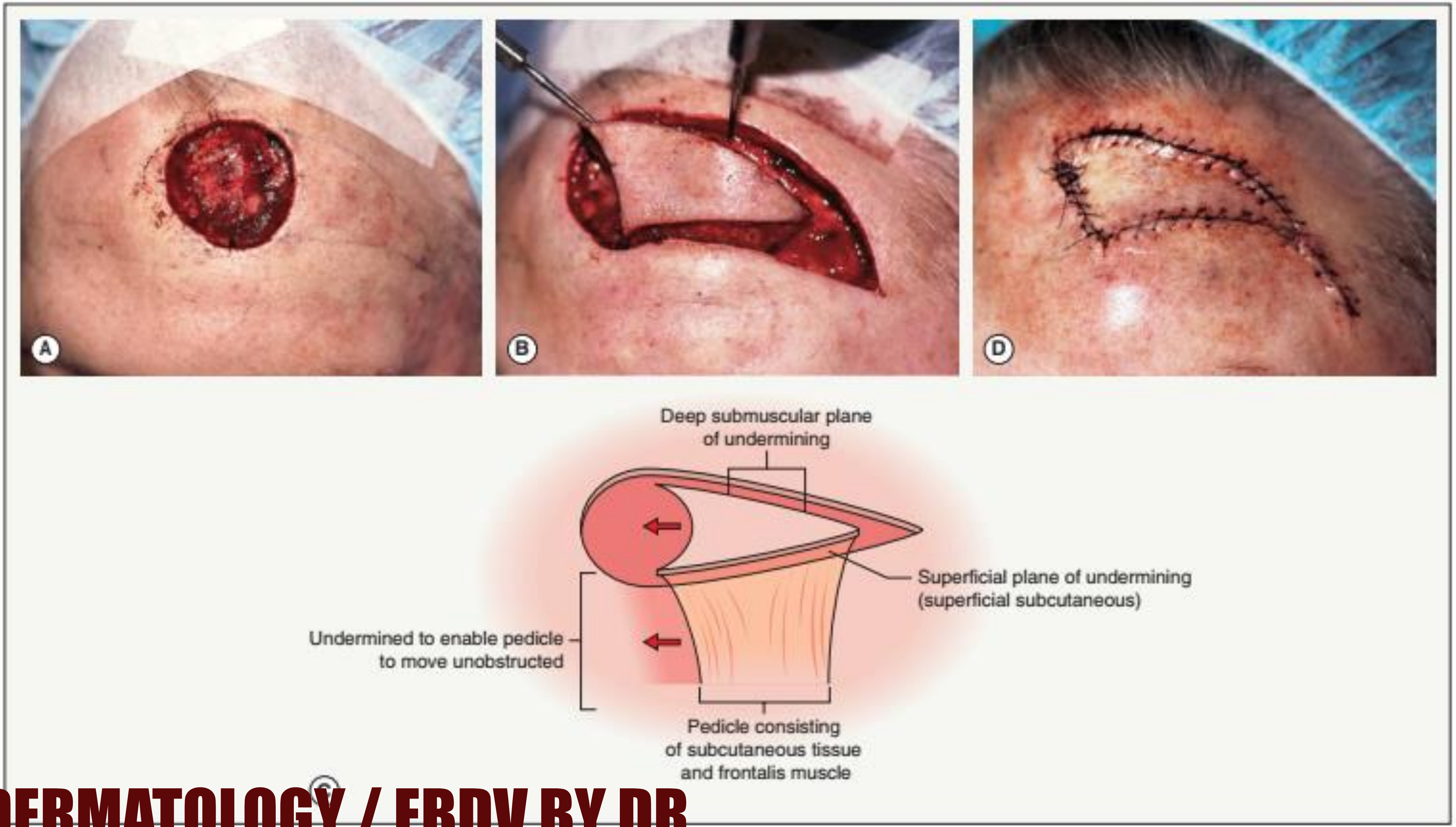
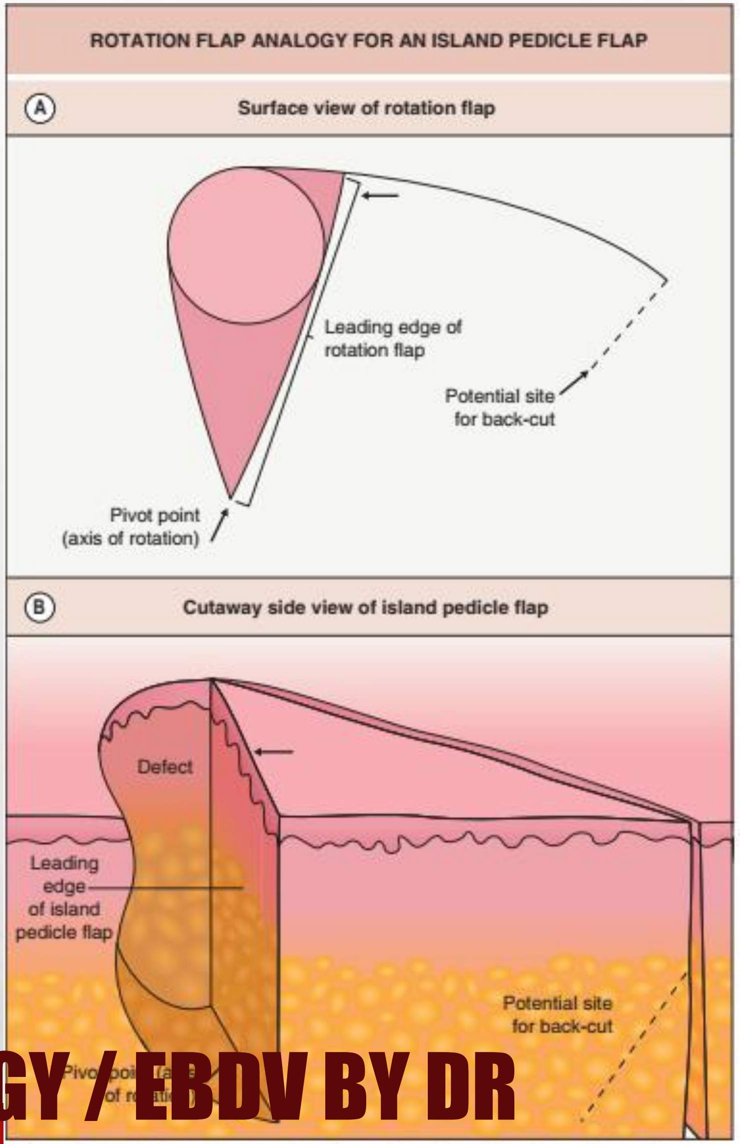
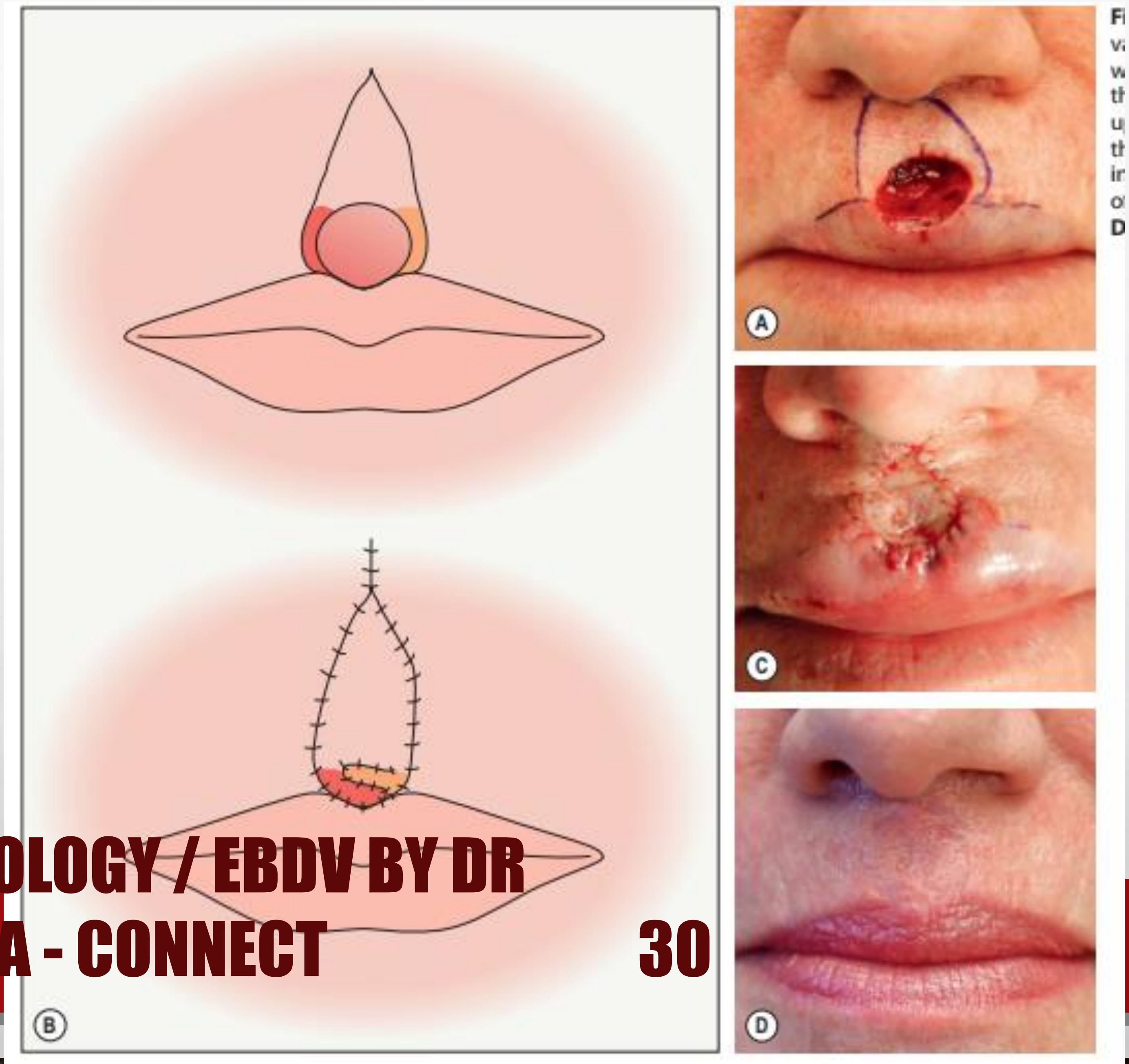


Fig. 147.20 Island pedicle flap. A Large mid-upper forehead defect. B Intraoperatively, an island pedicle flap is designed with a single inferior muscular pedicle [illegible] supp[illegible] and supraorbital arteries. C The fro[illegible] muscle serves as the inferiorly based pedicle for the flap. This [illegible] e subcutis beginning at the lower [illegible] of the flap (superficial plane) and undermining in the submuscular [illegible] the superior edge of the flap (deep plane). The skin medial to the pedicle must also be undermined to facilitate its advancement in the direction of the defect. D The defect is now comprised of acute angles which are more easily closed primarily.



...alogy for an island pedicle flap. **A** A rotati... ...urow's triangle adjacent to the defect and th... ...h would facilitate flap movement. **B** Side vie... defect and an island pedicle flap. The plane of rotation is perpendicular to the skin surface. The shaded area is where the equivalent of a Burow's triangle has been removed to facilitate flap movement. The dotted line is analogous to the back-cut of a rotation flap. Note the similarities between a rotation flap and an island pedicle flap.


A
B
C
D

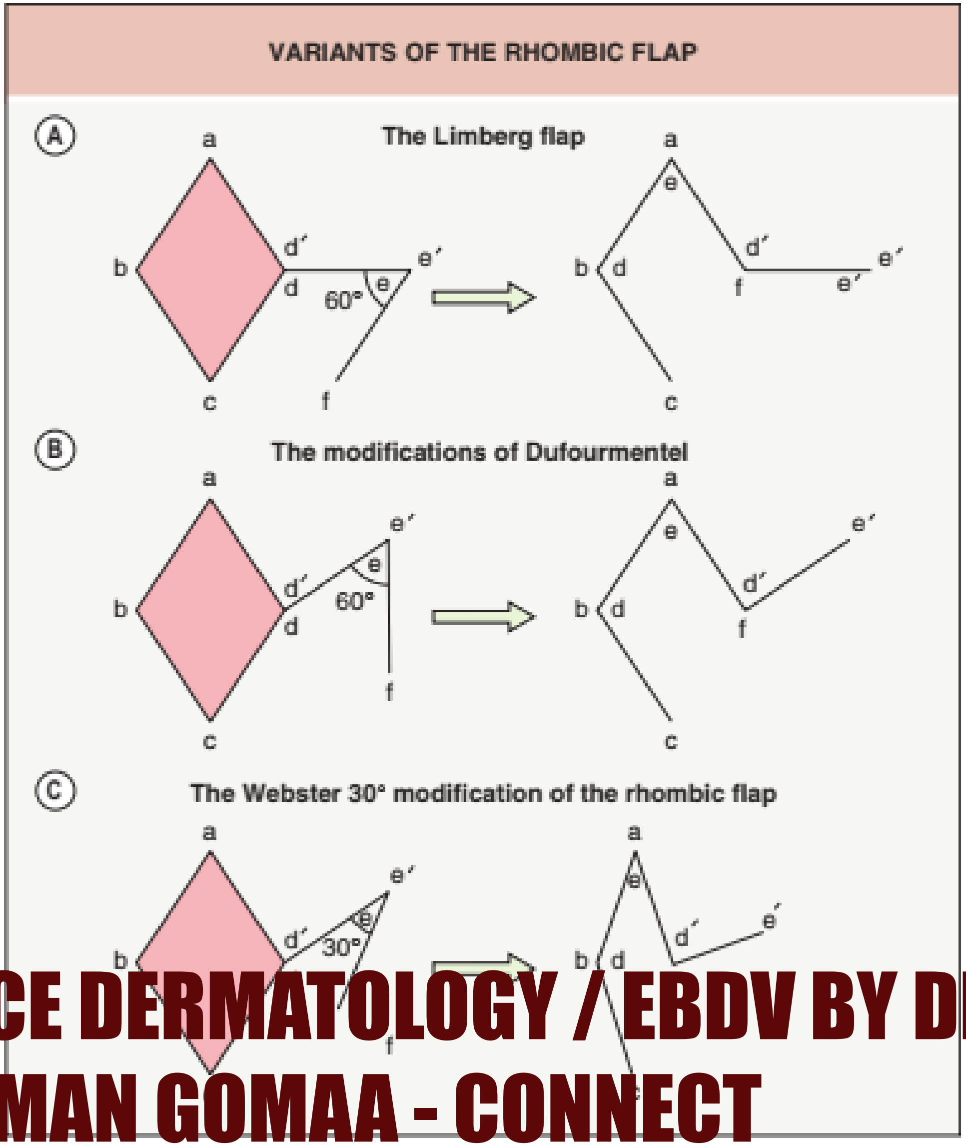


Fig. 147.23 Variants of the rhombic flap. **A** The Limberg flap is the standard [illegible] modifications of Dufourmentel. **C** The Webster 30° [illegible] rhombic flap.

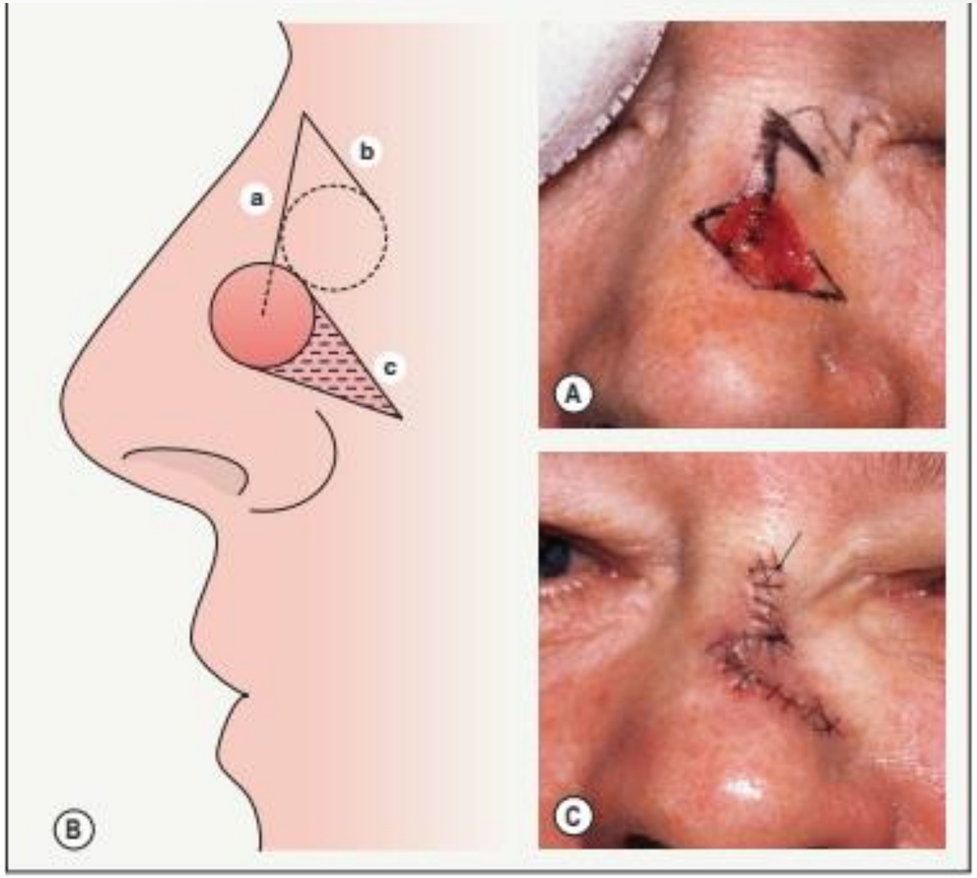


Fig. 147.24 Rhombic transposition flap on the upper lateral nose. A,B Key to the correct design of this flap is identification of the center point of the defect. The first incision for the flap is made in a line (**a**) along a vector radiating from the central point into the donor skin. The second incision (**b**) is made at an angle between 45° and 60° to this radiating incision, keeping in mind that the width of the flap must equal the width of the defect. Next, a 30° Burow's tria[illegible] excised such that the flap side of the incision (**c**) is parallel to the late[illegible]ge of the rhombic flap (**b**). **C** The tip of the flap is trimmed to fit the defect and the wound edges are sutured.

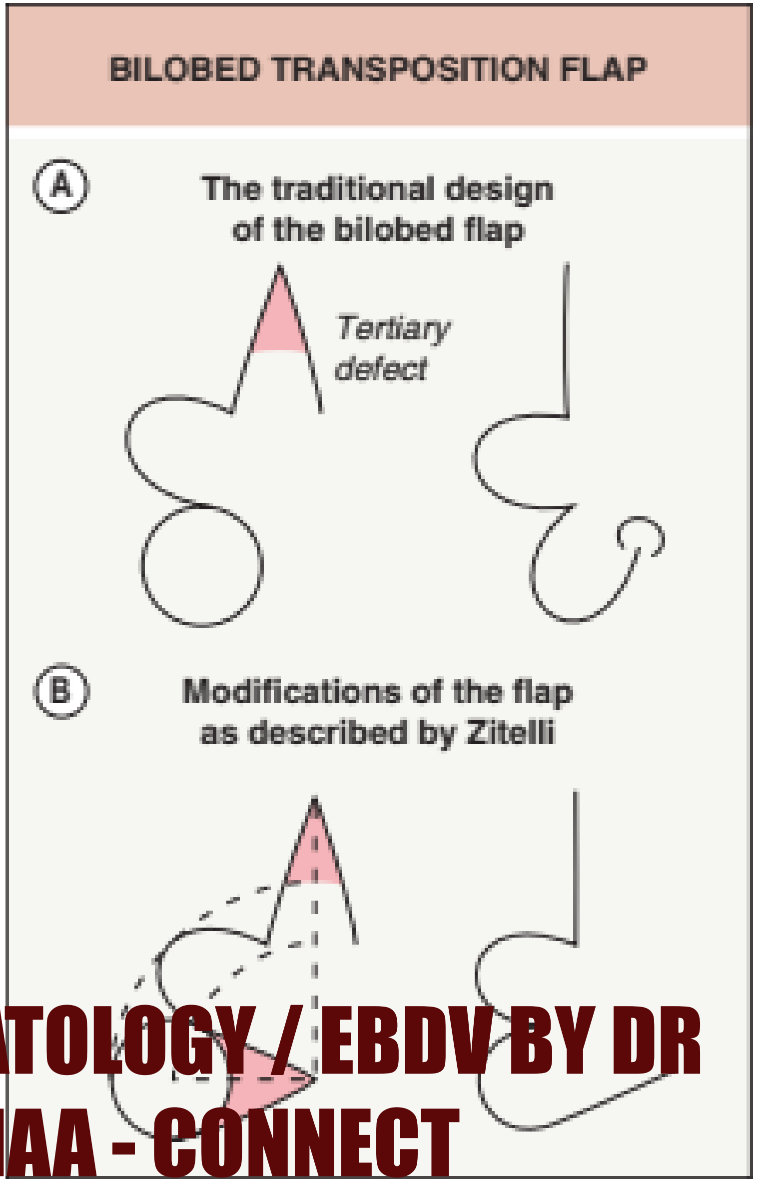


Fig. 147.25 Bilobed transposition flap. A The traditional design of the bilobed flap results in tissue protrusion at the pivot point. **B** Modifications of the bilobed flap as described by Zitelli.

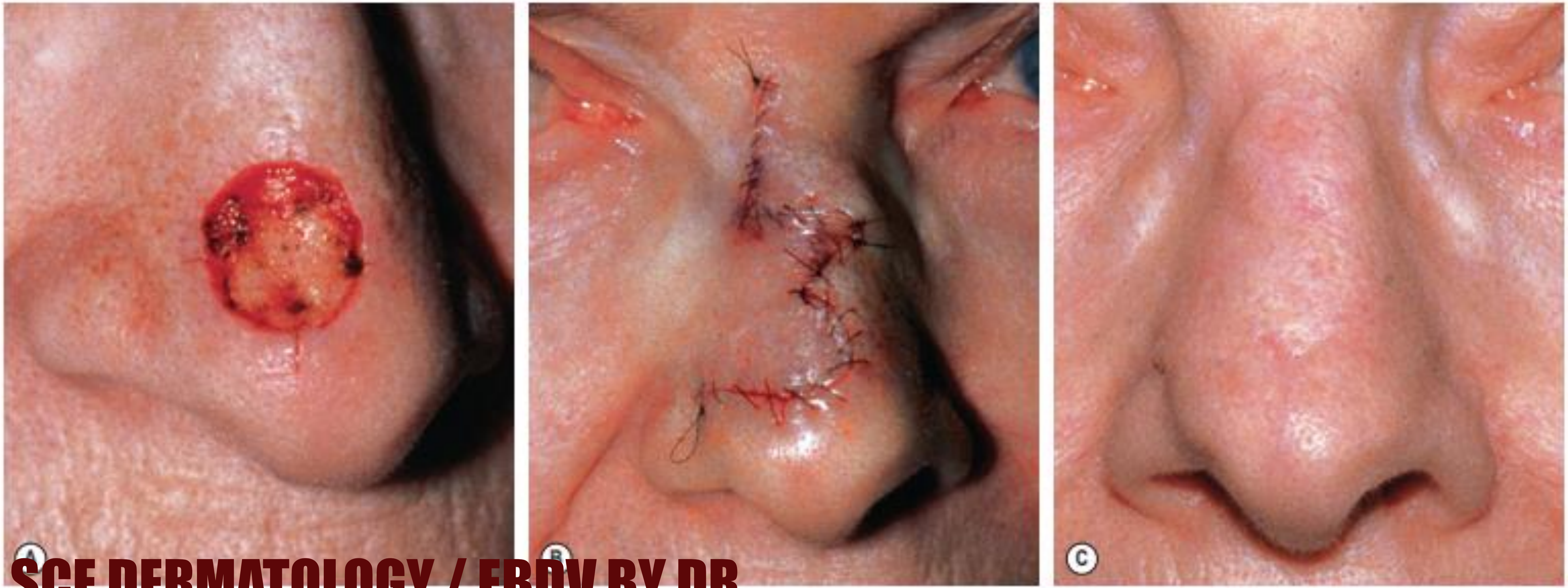
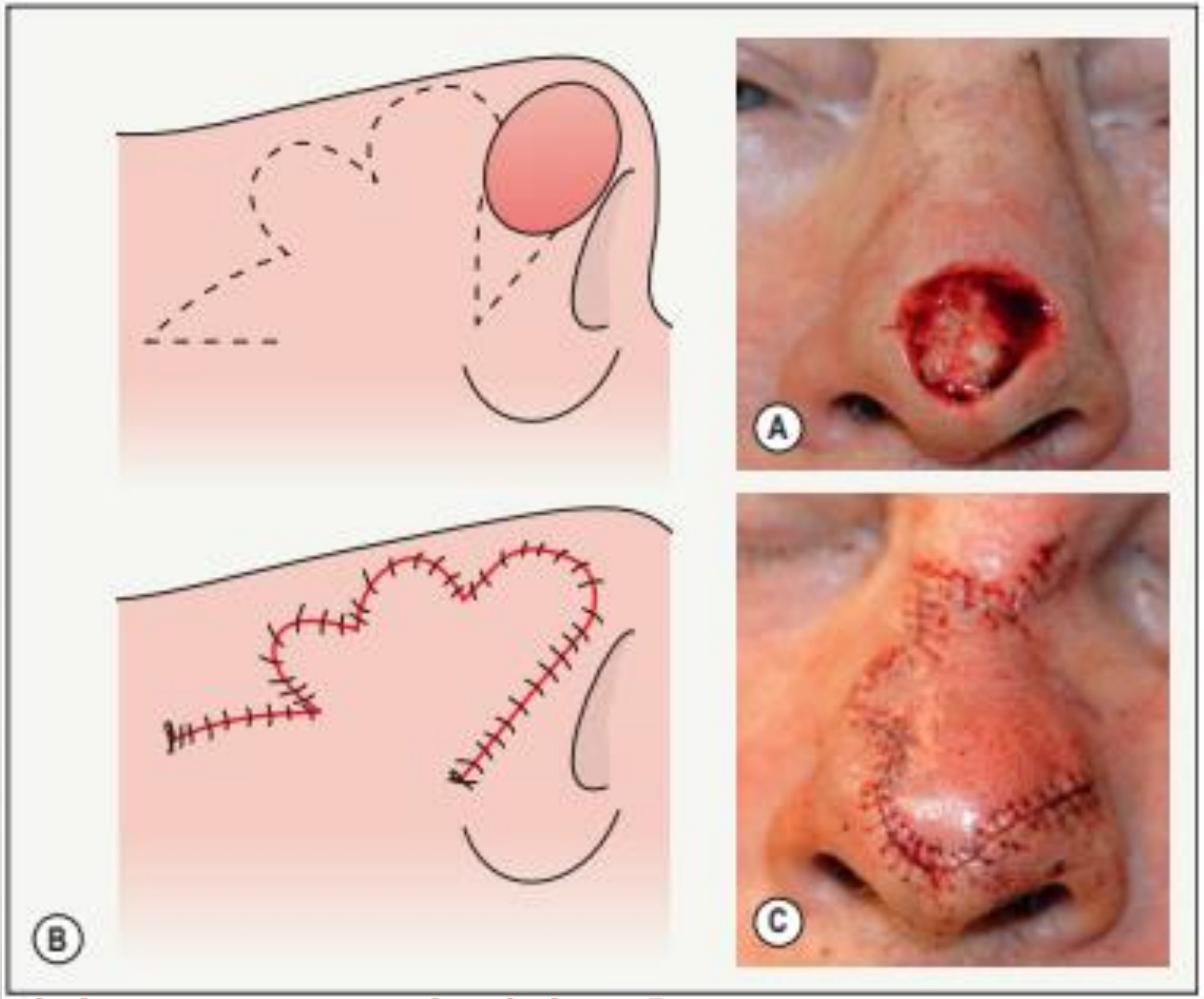


Fig. 147.26 Bilobed flap. A Distal nose defect. B Defect reconstructed with a modified bilobed transposition flap. C Cosmetic result at 1 year postoperatively.



F[illegible] flap with tertiary lobe oriented perpendicular to the alar rim. These three lobes ar[illegible] ects. A total of five Z[illegible] ng nearly tension-free redistribution of skin from the upper nose to the nasal tip. C Closed wound w[illegible] on of the nasal tip and alar structures.

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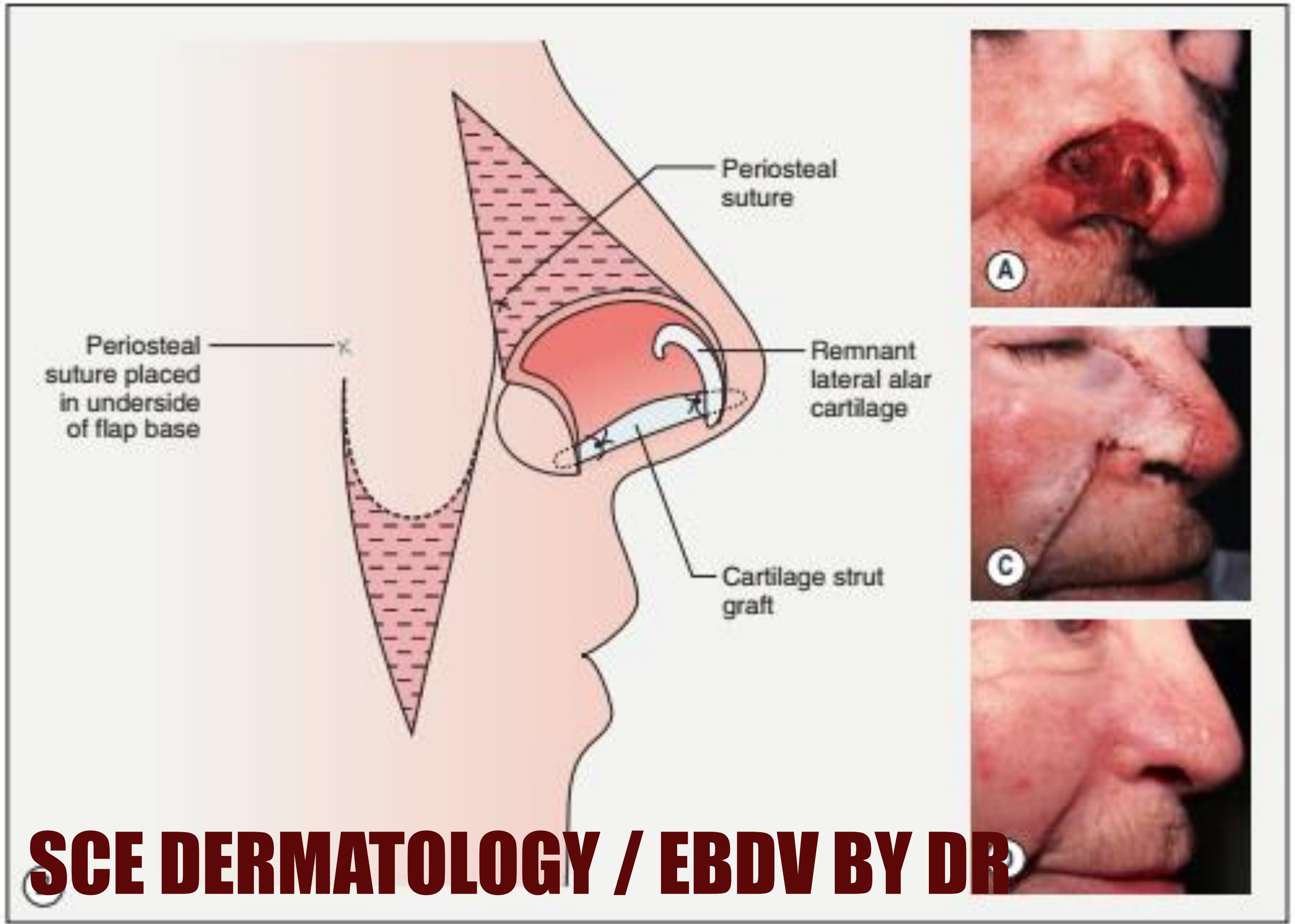

Fig. 147.28 Nasolabial transposition flap. A Large nasal defect involving loss of cartilage and extending to the intranasal skin. **B** Excision of a Burow's triangle on the nasal sidewall is oriented towards the medial canthal tendon. The lateral incision of the flap should extend no higher than the apex of alar skin over which the flap is to be transposed. In this case, an auricular cartilage strut graft is placed along the alar rim and inserted into medial and lateral dermal pockets and anchored with deep sutures into the wound base. The points of placement of the periosteal sutures into the undersurface of the advancing flap are indicated with an X. The distal flap is aggressively defatted. **C** The flap is trimmed to fit the defect, wrapped around the alar rim and folded onto itself, reforming the rim. **D** Cosmetic result at 6 months postoperatively.

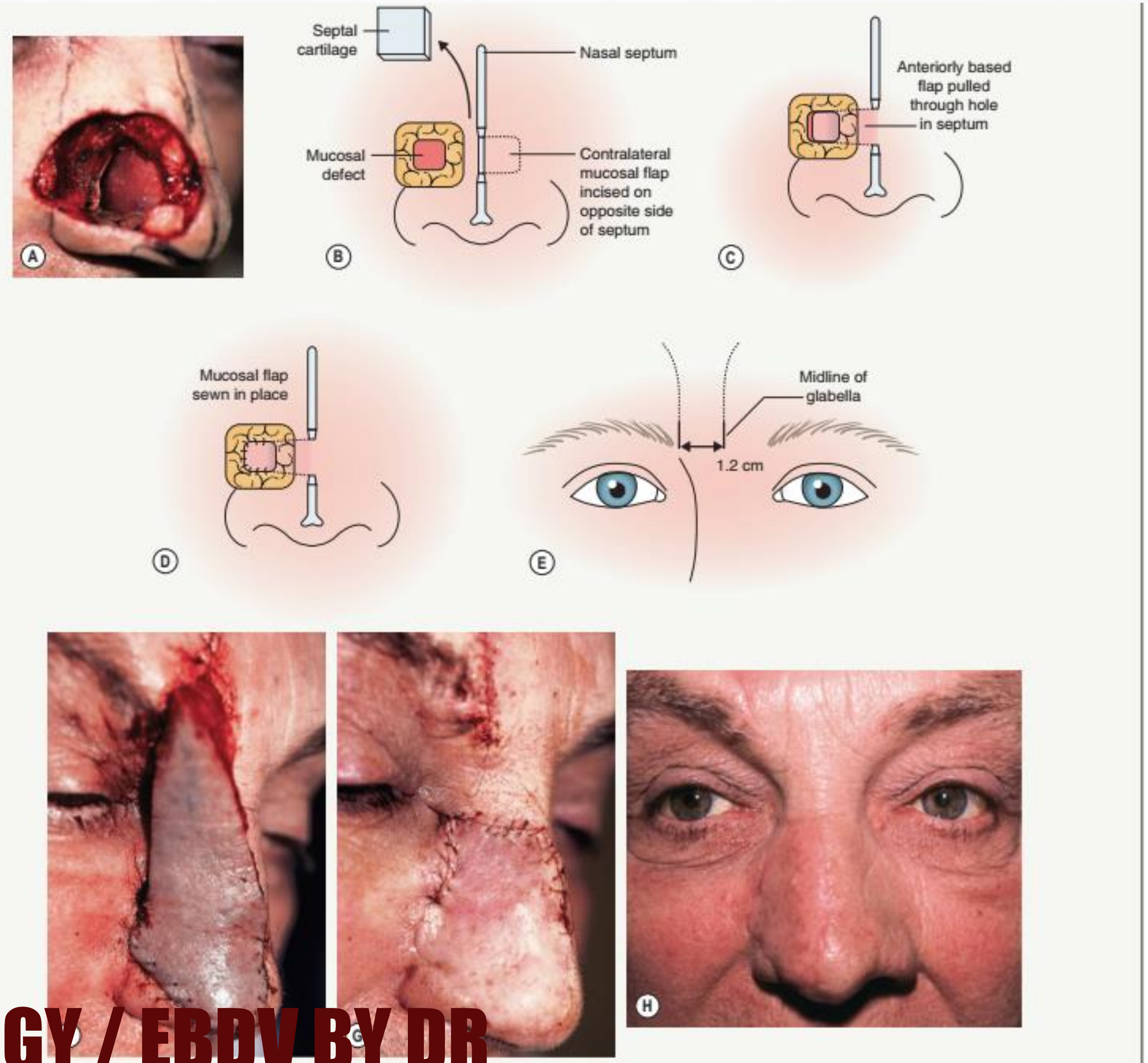

Fig. 147.29 Paramidline forehead flap. A Large full-thickness defect of the right half of the distal nose. B The mucosal defect is reconstructed with a contralateral [illegible]ptal mu[illegible] [illegible]ncision is made in the contralateral septal muc[illegible] anteriorly and correctly sized to cover the defect. In [illegible]dition, [illegible] [illegible]rvest[illegible] from the nasal septum and used to reconstitute [illegible] [illegible]tural integrity of the nasal sidewall. (Note that the [illegible]mainder [illegible] [illegible]unit and the right nasal tip subunit will be excise[illegible] reconstruction with the forehead flap.) C The mucosal flap is delivered through the septal defect into the right nasal vestibule. D The mucosal flap is sutured into place. E The design of the pedicle uses the midline of the glabella as the medial incision point; the lateral incision is made 1.2 cm lateral to the midline. Using a template, the forehead flap is then incised and raised on this 1.2-cm-wide pedicle. F Immediately postoperatively. G Immediately after takedown of flap 3 weeks later. H Early cosmetic results with no revision 2 months

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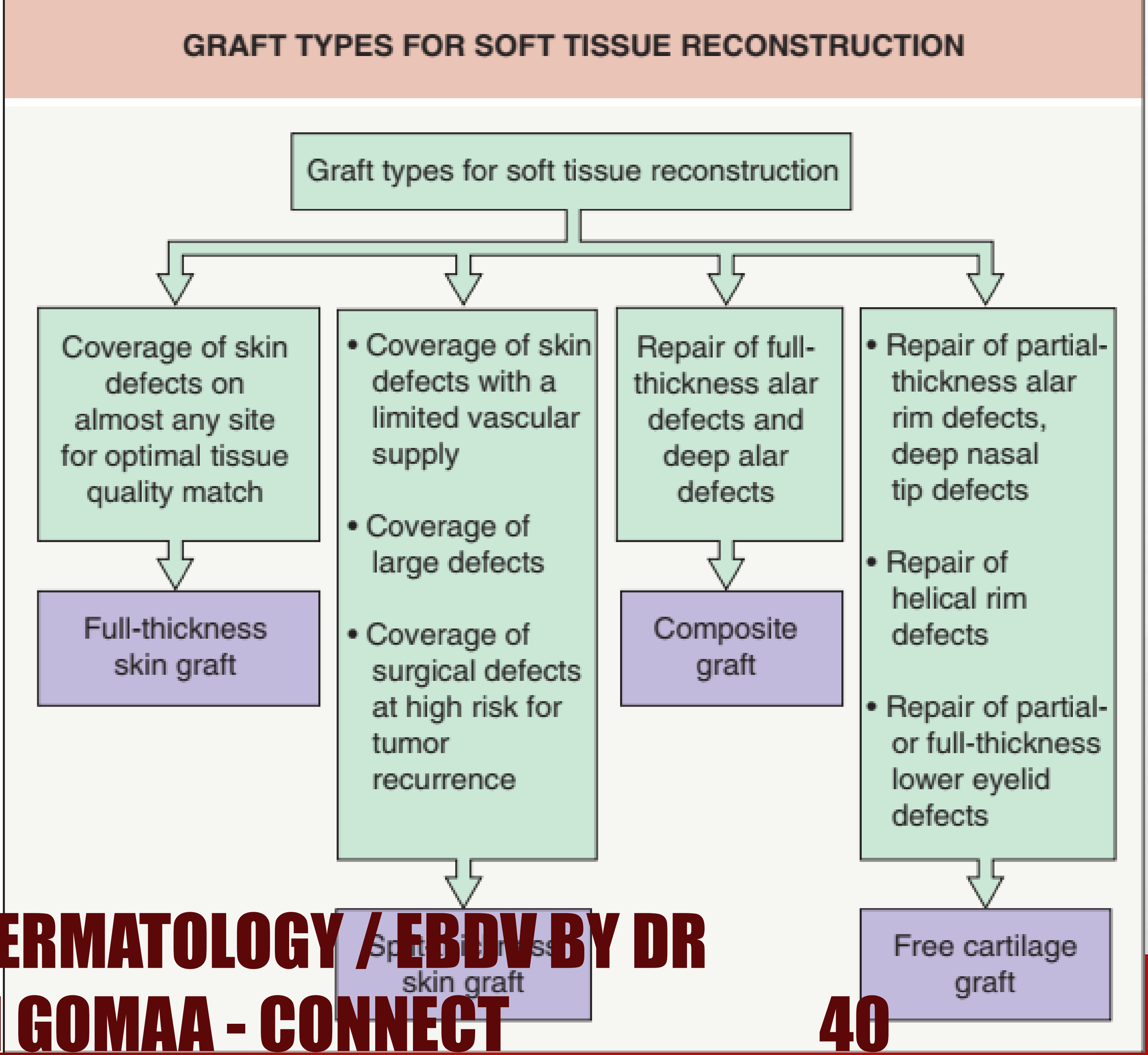
Fig. 147.30 Nasolabial interpolation flap. A Large alar defect with severe loss of structural integrity. **B** Nasolabial interpolation flap in place. A cartilage strut from the conchal bowl is sutured into the wound bed parallel to the alar rim to restore structural integrity. **C** Preoperatively (3 weeks after initial reconstruction), before takedown of the pedicle. **D** Postoperatively, at time of transection of pedicle. **E** Eighteen months postoperatively.

COMMON LOCATIONS FOR FLAP TYPES
Burow's triangle displacement flaps
Burow's flap (single tangent advancement flap) and bilateral single tangent advancement flap (A-to-T flap)
• Distal, lateral nasal sidewall • Upper lateral lip • Infranasal upper lip • Eyebrow and suprabrow forehead • Mid helix
Unilateral or bilateral double tangent advancement flap (U- or H-flap)
• Forehead • Helix • Large defect of upper lip
Rotation flap(s)
• Upper cheek • Medial chin • Nasal dorsum (Rigor flap) • Lower eyelid
Defect reconfiguration flaps
Island pedicle flap
• Upper lateral and mid upper cutaneous lip • Mucosal lip • Lateral chin and lower lip • Distal nasal sidewall • Preauricular cheek • Eyebrow
Tissue reorientation flaps
Rhombic transposition flap
• Nasal sidewall • Temples and lateral forehead • Lateral canthus and lateral lower lid • Lateral chin and lower lip • Superior and anterior helix
Bilobed transposition flap
• Nasal tip and supratip • Lower lateral lip • Periocular skin
Nasolabial transposition flap
• Alar defects, especially those involving the lateral portion of the ala • Lateral alar crease and distal nasal sidewall
Tissue importation flaps
Paramedian/paramidline forehead flap
• Large nasal defects • Moderate-size nasal tip defect • Nasal defects with loss of cartilage and/or nasal lining • Large medial canthus defects
Nasolabial (melolabial) interpolation flap
• Nasal ala and alar rim • Nasal tip • Mid upper lip
Melolabial flap with a twist (Spear's flap)
• Full-thickness alar defects involving the junction of the lateral ala and the cheek/upper lip
Retroauricular flap
• Large defects of the anterior ear or helix • Full-thickness defects requiring cartilage graft reconstruction

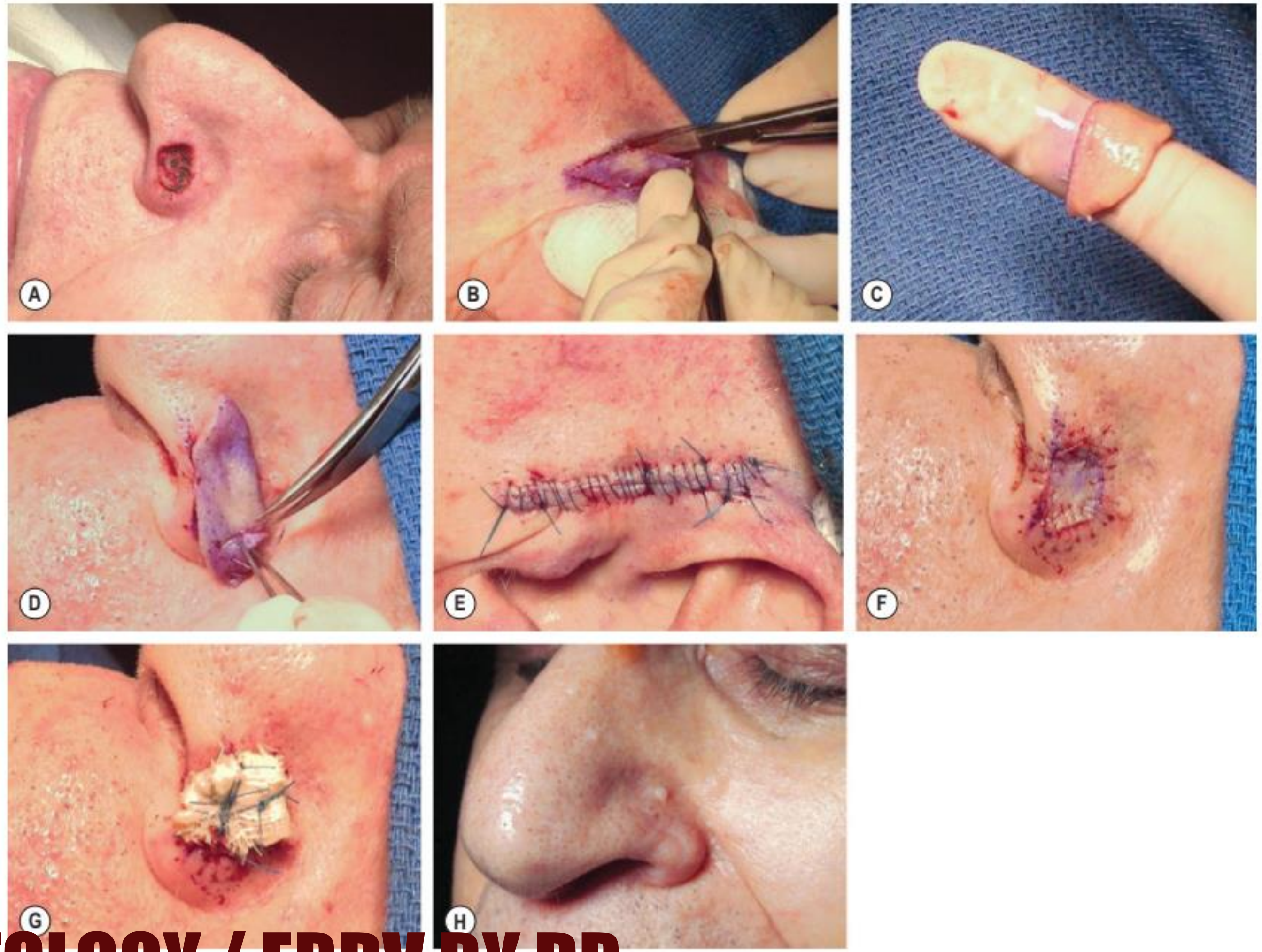
GRAFTS


GRAFT TYPES FOR SOFT TISSUE RECONSTRUCTION
Graft types for soft tissue reconstruction
Coverage of skin defects on almost any site for optimal tissue quality match
Full-thickness skin graft
• Coverage of skin defects with a limited vascular supply
• Coverage of large defects
• Coverage of surgical defects at high risk for tumor recurrence
skin graft
Repair of full-thickness alar defects and deep alar defects
Composite graft
• Repair of partial-thickness alar rim defects, deep nasal tip defects
• Repair of helical rim defects
• Repair of partial- or full-thickness lower eyelid defects
Free cartilage graft

COMPARISON OF GRAFT TYPES USED IN SOFT TISSUE RECONSTRUCTION								
Graft type	Tissue match	Nutritional requirements	Requirement for recipient bed vascularity	Infection risk	Graft contraction risk	Durability	Sensation	Adnexal function
FTSG	Good to excellent	High	High	Low	Low	Good to excellent	Good	Excellent
STSG	Poor to fair	Low	Low	Low	High	Fair to good	Fair	Poor
Composite	Good	High	Very high	Moderate	Low	Fair	Fair	Good
Free cartilage	NA	Moderate	High	Moderate	Migration or deformation possible, with subsequent resorption	Good	NA	NA

INDICATIONS, CONTRAINDICATIONS, ADVANTAGES AND DISADVANTAGES OF GRAFT TYPES USED IN SOFT TISSUE RECONSTRUCTION				
Graft type	**Indications**	**Contraindications**	**Advantages**	**Disadvantages**
Full-thickness skin graft (FTSG)	Coverage of skin defects, usually located on the face or scalp, requiring an optimal tissue quality match	Avascular graft bed	Excellent color and texture match Minimal graft contraction	Time consuming Donor site requires closure Requires meticulous technique
Split-thickness skin graft (STSG)	Coverage of large defects and defects with a limited vascular supply or at sites at high risk for tumor recurrence	Defects near free margins Facial defects in cosmetically sensitive areas	Increased chance of survival Ease of application Ability to cover large defects Able to act as a "window" to detect recurrence of high-risk tumors	Suboptimal cosmetic appearance Decreased durability Presence of granulating donor site wound Greater graft contraction Special equipment required to harvest larger grafts
Perichondrial cutaneous graft	Repair of deep nasal tip and alar defects, especially those with exposed cartilage	Avascular graft bed	Thicker than FTSG Contract less than FTSG Increased chance of survival under conditions of vascular compromise (compared to FTSG)	
Composite graft	Repair of full-thickness alar defects or nasal tip defects (<2 cm in diameter) with cartilage loss	Defects >2 cm in diameter	Single-stage procedure May yield good functional and aesthetic results when used to repair appropriately sized defects	Greater risk of graft failure under conditions of vascular compromise Graft size limitations (<2 cm) Limited donor tissue available Donor site may require flap or complex closure Increased risk of infection and graft displacement
Free cartilage graft	Repair of defects of nasal ala, tip or sidewall [illegible] requiring restoration of structural support or preservation of contour of free margins	None	Provide structural support for selected facial and ear defects	Donor site may be painful Increased risk of postoperative infection Increased risk of graft displacement

Fig. 148.2 Full-thickness skin graft. A Full-thickness skin graft taken from the preauricular region was used to repair this defect of the right upper anterior ear. **B** Eight-week postoperative result.



[illegible] Placement of a full-thickness skin graft. A Defect [illegible] left [illegible], measuring 1.8×1.2 cm, is present after removal of basal cell carcinoma via Mohs micrographic surgery. A preauricular donor site will provide the best match based upon color, sebaceous quality, degree of photodamage, and thickness. B With a marking pen, a template has been made from the periphery of the recipient site and the template is then used to produce an inked margin at the donor site [illegible] ext, B[illegible] on either side of the template. Marking [illegible] or site prior to injecting local anesthesia prevents incorrect [illegible] tretch[illegible]. The donor site is excised with a scalpel t[illegible]el of the subcutaneous fat. C The graft is carefully defatted with curved iris scissors so that only the white, glistening surface of the dermis remains. D The full-thickness skin graft is trimmed with curved iris scissors to ensure a perfect fit. E The preauricular donor site is closed primarily. F Full-thickness skin graft sewn into place with 6-0 fast-absorbing gut sutures. G Xeroform™ bolster [illegible] into place over the full-thickness skin graft with 5-0 polypropylene tie-over sutures. H Sixteen-week postoperative result. The graft shows good color and [illegible]re match with the surrounding skin.

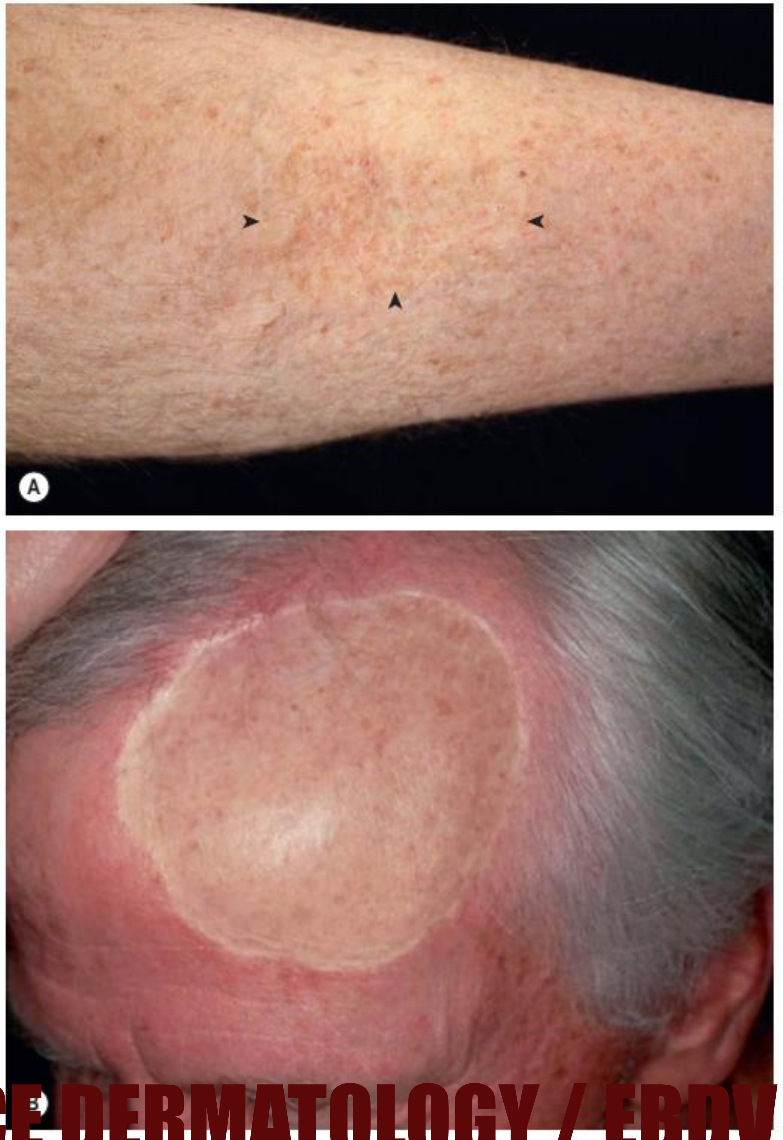


Fig. 148.4 Split-thickness skin grafts (STSGs). A A relatively thick STSG placed on the medial aspect of the lower extremity with excellent cosmetic results: [illegible] graft. B [illegible] and [illegible] Note the relative hypopigmentation, altered texture, and depression of the STSG as compared to the normal surrounding skin. *Courtesy, Jean L Bolognia, MD.*

CLASSIFICATION OF SPLIT-THICKNESS SKIN GRAFTS	
Thin	0.005–0.012 in/0.13–0.30 mm
Medium	0.012–0.018 in/0.30–0.46 mm
Thick	0.018–0.030 in/0.46–0.76 mm

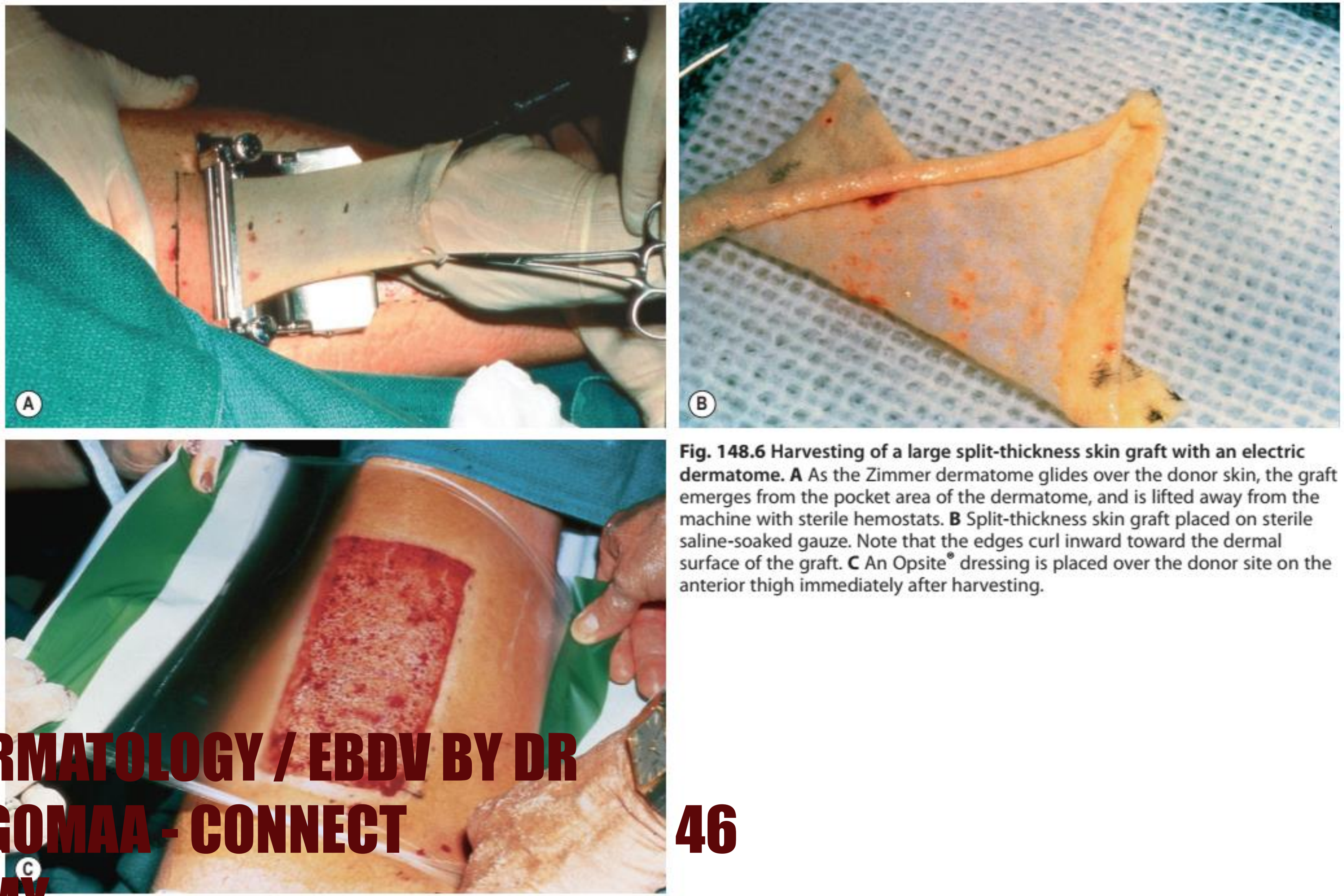
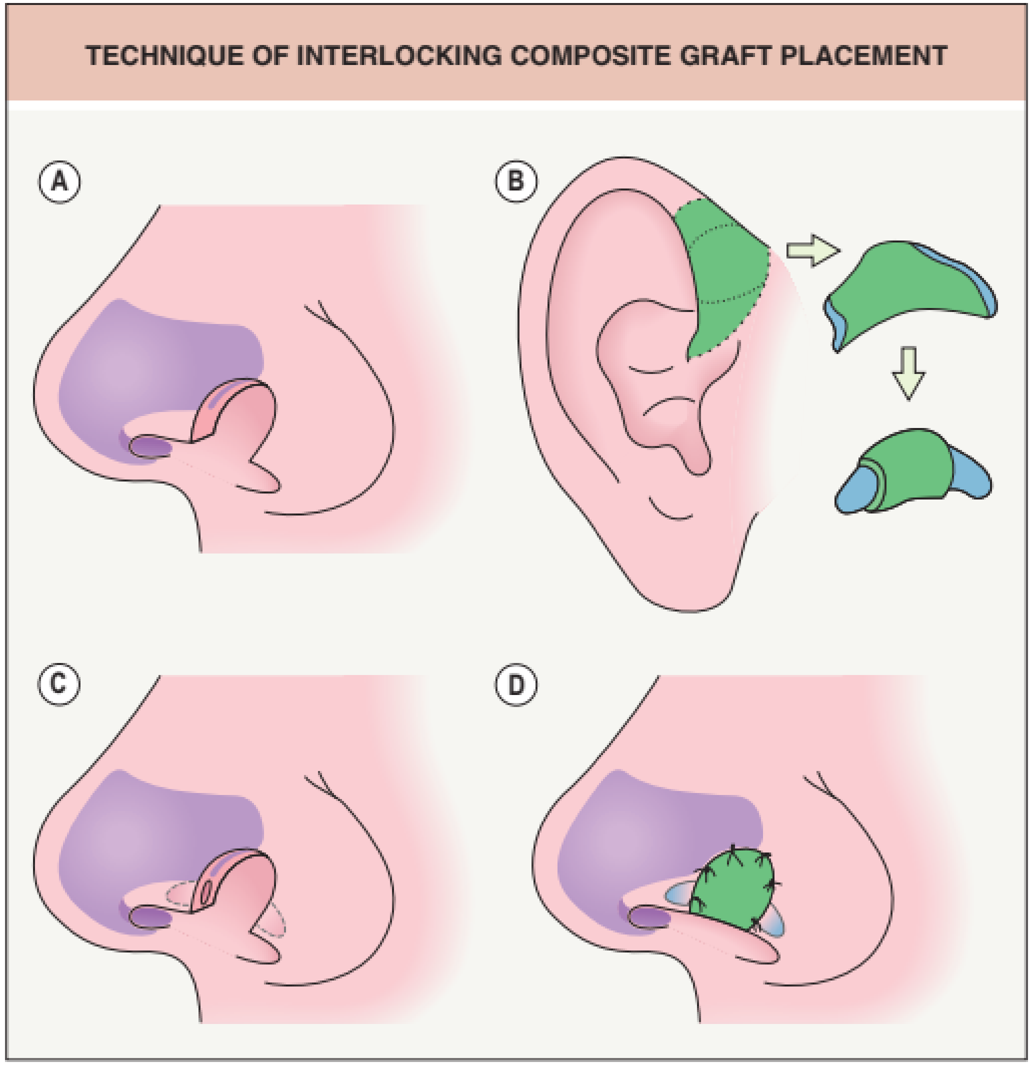


Fig. 148.6 Harvesting of a large split-thickness skin graft with an electric dermatome. A As the Zimmer dermatome glides over the donor skin, the graft emerges from the pocket area of the dermatome, and is lifted away from the machine with sterile hemostats. **B** Split-thickness skin graft placed on sterile saline-soaked gauze. Note that the edges curl inward toward the dermal surface of the graft. **C** An Opsite® dressing is placed over the donor site on the anterior thigh immediately after harvesting.



requires repair. **B** Donor site at the crus of the helix. An area of skin ~5–10%

both sides to expose the cartilaginous portion of the graft, such that two

which the cartilaginous pegs will be placed. **D** The cartilaginous pegs are

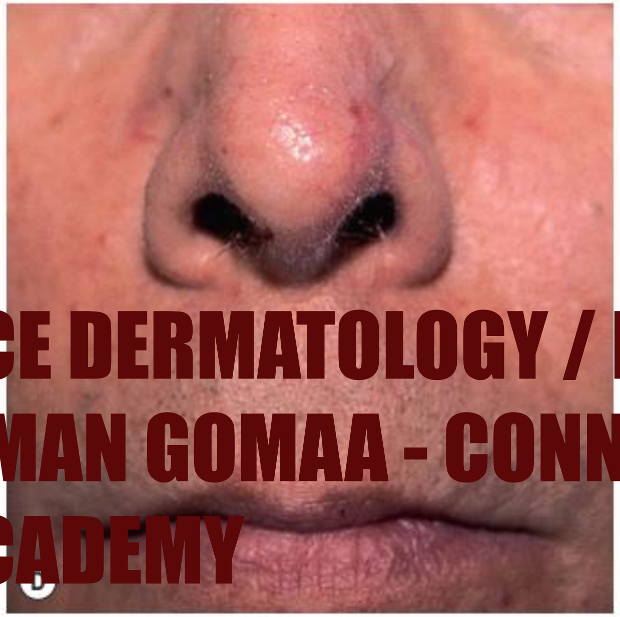


Fig. 148.10 Harvesting and placement of a free cartilage graft. A The posterior conchal bowl donor site has been incised to the level of the perichondrium. The desired length of cartilage was incised with the scalpel, and a second incision made parallel to the first to isolate a cartilaginous strip with its overlying perichondrium. **B** The ends of the cartilaginous strip with its overlying perichondrium have been inserted into pockets undermined on either side of the recipient bed, such that the graft interlocks with its recipient bed on the left nasal ala. The graft is sutured to the underlying dermis with one 5-0 absorbable suture for additional security. **C** A full-thickness skin graft has been sutured into place over the free cartilage graft. **D** Eight-week postoperative view of the full-thickness skin graft and underlying free cartilage graft on the left nasal ala. The alar rim remains in perfect alignment.

CAUSES OF GRAFT FAILURE	
Poor graft–bed contact	Hematoma Seroma Shearing forces due to excessive postoperative activity Shearing forces due to inadequate immobilization of graft
Poor recipient bed vascularity	Inadequate vascular bed (e.g. previous radiotherapy) Exposed cartilage, bone, tendon Nicotine-induced vasoconstriction
Infection	Coagulase-positive *Staphylococcus* spp. Beta-hemolytic *Streptococcus* spp. *Pseudomonas* spp.
Host factors	Diabetes mellitus Immunosuppression Poor general health Nutritional deficiencies, including hypoalbuminemia Systemic conditions with vascular compromise
Technique	Rough handling of tissue Excessive devitalized tissue in recipient bed due to cautery Increased [illegible] on graft [illegible] inadequate size Inadequate hemostasis Inadequate [illegible] layer [illegible] graft (more common with STSG) Inadequate anchoring (composite and free cartilage grafts)

DONOR SITE CONSIDERATIONS FOR FULL-THICKNESS SKIN GRAFTS	
• Pattern of sun exposure • Color • Vascular pattern • Texture • Sebaceous quality	• Skin thickness • Skin consistency • Sufficient tissue available for harvesting • Ability to camouflage donor site scar

SUMMARY OF POSSIBLE FULL-THICKNESS SKIN GRAFT DONOR SITES FOR DEFECTS IN DIFFERENT LOCATIONS	
Defect sites	**Donor sites**
Nasal dorsum, sidewalls, tip	Preauricular region, supraclavicular region or lateral neck (if large)
Nasal tip, ala	Preauricular region, conchal bowl, nasolabial fold
Junction of nasal dorsum/ tip	Burow's graft
Ear	Postauricular sulcus, preauricular cheek
Lower eyelid/medial canthus	Upper eyelid, postauricular sulcus
Scalp	Supraclavicular region, lateral neck, inner upper arm
Forehead	Burow's graft, supraclavicular region, lateral neck, inner upper arm (if large)

Table 6-8: Stages of Skin Graft Survival

Stages	Description
1. Imbibition	First 24-48 hours (ischemic period) Graft sustained by plasma exudate from wound bed Fibrin attaches graft to new bed Graft becomes edematous, ↑ weight by up to 40%
2. Inosculation	Begins 48-72 hours, lasts up to 7-10 days (graft vessels anastomose) Revascularization linking dermal vessels in graft to recipient bed Rationale for delayed grafting over sites devoid of perichondrium or periosteum (allows formation of granulation tissue with ↑ survival rate)
3. Neovascularization	Occurs temporally with inosculation Capillary ingrowth from recipient wound base and sidewalls to graft If optimal conditions, full circulation reestablished within 4-7 days
4. Maturation	Occurs months later Reinnervation of graft typically within 2 months of graft but may not be complete for months to years (full sensation may never fully return)


THANK
YOU